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PHYSICS

1) **Assertion :** The apparent weight of a person standing in a lift which moves upwards with uniform acceleration is always higher than his true weight.
Reason : The weight (on the earth surface) always acts downwards.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

2) **Assertion (A) :** Frictional force is the component of contact force parallel to the surface.
Reason (R) : Friction force always opposes the motion of a body.
In the light of the above statements, choose the most appropriate answer from the options given below :

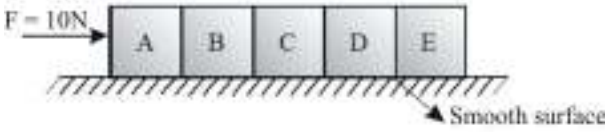
(1) **(A)** is false but **(R)** is true.

(2) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(3) Both **(A)** and **(R)** are true and **(R)** is not the correct explanation of **(A)**

(4) **(A)** is true but **(R)** is false.

3) As shown in diagram, there are 5 blocks each of the mass 1 kg resting on a smooth horizontal surface.



Match the columns and select the correct option :

Column I		Column II	
A.	Acceleration of system is (in m/s^2)	**P.**	4
B.	Normal force between B and C is (in N)	**Q.**	2
C.	Normal Force between A and B is (in N)	**R.**	6
D.	Normal Force between C and D is (in N)	**S.**	8

(1) A-Q; B-R; C-S; D-P

(2) A-S; B-R; C-Q; D-P

(3) A-P; B-Q; C-R; D-S

(4) A-R; B-S; C-P; D-Q

4) **Assertion :** Mass is a measure of inertia of the body in linear motion.
Reason : Greater the mass, greater is the force required to change its state of rest or of uniform motion.

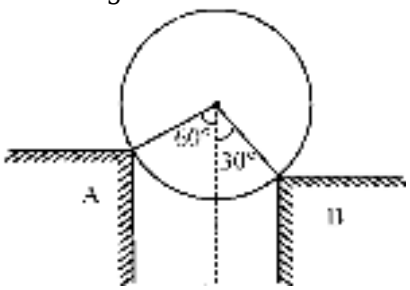
(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(2) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(3) Assertion is true but Reason is false.

(4) Both Assertion and Reason are false.

5) A smooth cylinder of mass m and radius R is resting on two corner edges A and B as shown in fig.



The relation between normal reaction at the edges A and B is :-

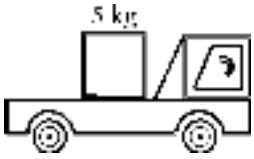
(1) $N_A = \sqrt{2}N_B$

(2) $N_B = \frac{2\sqrt{3}N_A}{5}$

(3) $N_A = \frac{N_B}{2}$

(4) $N_B = \sqrt{3}N_A$

6) What will be the maximum acceleration of the truck so that the block of mass 5 kg placed inside the truck does not slip ? If the cofficient of friction $\mu = 0.3$:-



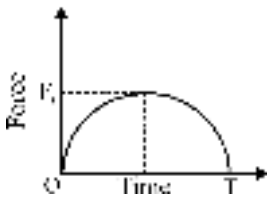
(1) 3 m/s^2

(2) 2 m/s^2

(3) 1 m/s^2

(4) Zero

7) A particle of mass m, initially at rest, is acted upon by a variable force F for a brief interval of time T. It begins to move with a velocity u after the force stops acting. F is shown in the graph as a function of time. The curve is a semicircle.



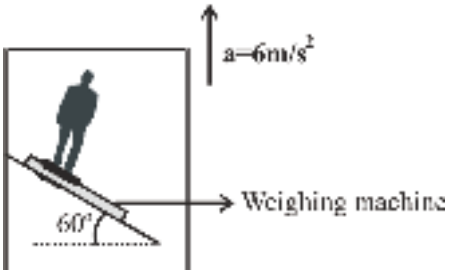
(1) $u = \frac{\pi F_0^2}{2m}$

(2) $u = \frac{\pi T^2}{8m}$

(3) $u = \frac{\pi F_0 T}{4m}$

(4) $u = \frac{F_0 T}{2m}$

8) An elevator is accelerating upwards with an acceleration of 6 m/s^2. Inside it a person of mass 50 kg is standing on a weighing machine which is kept on an inclined plane having angle of inclination 60°. The reading of the weighing machine is :



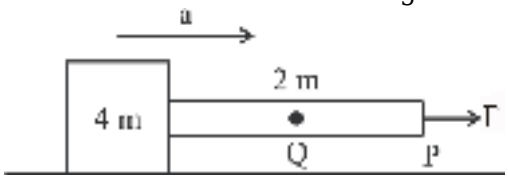
(1) 40 kg –Wt

(2) 160 kg –Wt

(3) 80 kg –Wt

(4) 50 kg –Wt

9) The system shown in figure is moving with constant acceleration. If force in the string at point P is F. Then tension in the string at mid point Q will be (mass of string 2m, mass of block is 4m) :-



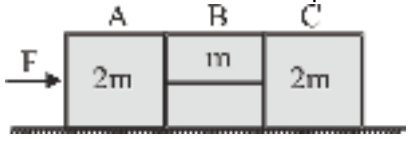
(1) $\frac{F}{4}$

(2) $\frac{5F}{6}$

(3) $\frac{3F}{4}$

(4) $\frac{F}{2}$

10) The system is pushed by a force F as shown in figure. All surfaces are smooth except between B and C. Friction coefficient between B and C is μ. Minimum value of F to prevent block B from downward slipping is :-



(1) $\left(\frac{3}{2\mu}\right) mg$

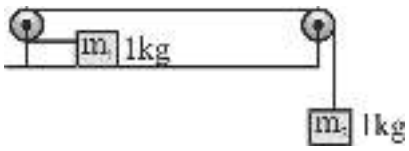
(2) $\left(\frac{5}{2\mu}\right) mg$

(3) $\left(\frac{5}{2}\right)\mu mg$

(4) $\left(\frac{3}{2}\right)\mu mg$

11)

Consider the system shown in figure. The pulley and the string are light and all the surfaces are frictionless. The tension in the string is



(take g = 10 m/s^2) :-

(1) 0 N

(2) 1 N

(3) 2 N

(4) 5 N

12) **Statement-1:** Frictional forces are conservative forces. **Statement-2:** Potential energy can be associated with frictional forces.

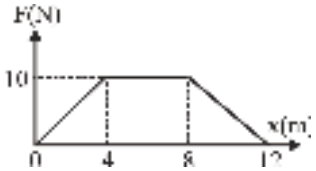
(1) Statement-1 is true, Statement-2 is true

(2) Statement-1 is false, Statement-2 is true

(3) Statement-1 is true, Statement-2 is false

(4) Statement-1 is false, Statement-2 is false

13) A particle of mass 0.1 kg is subjected to a force which varies with distance as shown in fig. If it starts its journey from rest at x = 0, its velocity at x = 12 m is :-



(1) 0 m/s

(2) $20\sqrt{2}$ m/s

(3) $20\sqrt{3}$ m/s

(4) 40 m/s

14)

Work done by static friction on an object:

(1) may be positive

(2) must be negative

(3) must be zero

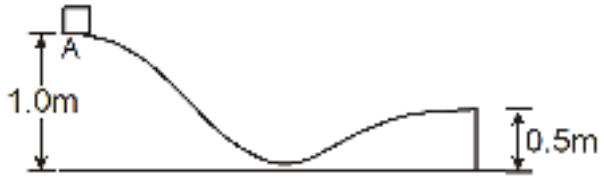
(4) none of these

15) A body of mass 1 kg is thrown upwards with a velocity 20 m/s. It momentarily comes to rest after

attaining a height of 18 m. How much energy is lost due to air friction ($g = 10\ m/s^2$)

(1) 30 J
(2) 40 J
(3) 10 J
(4) 20 J

16) Figure shows a particle sliding on a frictionless track which terminates in a straight horizontal section. If the particle starts slipping from the point A, how far away from the track will the particle hit the ground ?



(1) At a horizontal distance of 1 m from the end of the track.
(2) At a horizontal distance of 2 m from the end of the track.
(3) At a horizontal distance of 3 m from the end of the track.
(4) Insufficient information

17) A vehicle of mass m is moving on a rough horizontal road with kinetic energy E. if the coefficient of friction between the tyres and the road is μ, then the stopping distance is :

(1) $\frac{E}{\mu mg}$
(2) $\frac{E}{2\mu mg}$
(3) $\frac{2E}{\mu mg}$
(4) $\frac{E^2}{2\mu mg}$

18) Under the action of a force a 4kg body moves such that its position x in meters as a function of time t is given by $x = 3t^2$. Then work done by the force in first second is :

(1) 72 J
(2) 54 J
(3) 60 J
(4) 36 J

19) A body crosses the topmost point of a vertical circle with critical speed. What will be its centripetal acceleration when the string is horizontal :–

(1) g
(2) 2g
(3) 3g
(4) 6g

20) A particle starts from rest and attains an angular acceleration $\alpha = t^2 - 2t + 2$. (t in seconds), calculate angular speed after 2 seconds :-

(1) $\frac{4}{3}$rad/sec

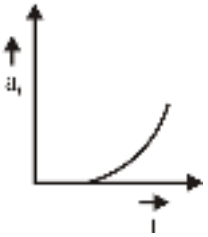
(2) $\frac{2}{3}$rad/sec

(3) $\frac{1}{3}$rad/sec

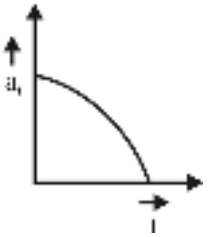
(4) $\frac{8}{3}$rad/sec

21) The angle turned by a body under going circular motion depend on time as $\theta = \theta_0 + \theta_1 t + \theta_2 t_0^2$ then draw the a_t v/s t (tangent acceleration v/s time graph for the body. (Radius = constant)

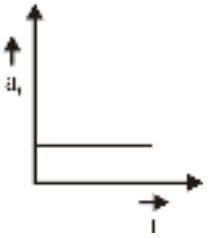
(1)



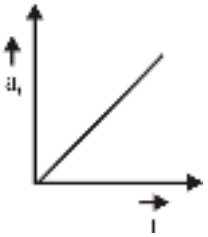
(2)



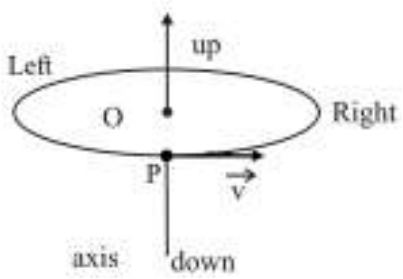
(3)



(4)



22) Velocity is shown for a body P. Find the direction of angular velocity of P?



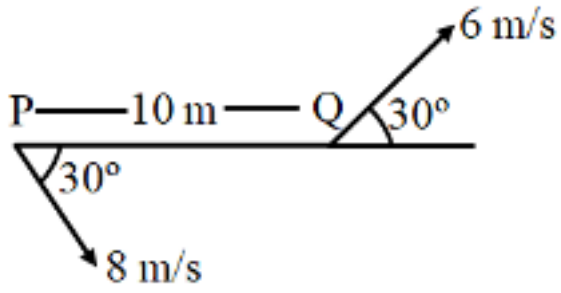
(1) left

(2) Right

(3) up

(4) down

23) Two moving particles 'P' and 'Q' are 10 m apart at an certain instant. The velocity of P is 8 m/s making an angle 30° with line joining P and Q and velocity of Q is 6 m/s making an angle 30° with PQ as shown in figure. Then the angular speed of P with respect to Q is -



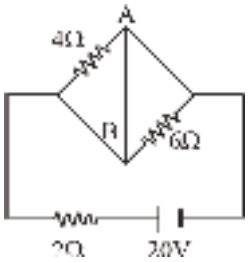
(1) 0 rad/s

(2) 0.1 rad/s

(3) 0.4 rad/s

(4) 0.7 rad/s

24) A circuit is shown in figure. Identify the **incorrect** statement among the following.



(1) The power supplied by the battery is 200W.

(2) The current flowing in the circuit is 10A

(3) The potential difference across 4Ω resistance is equal to the potential difference across 6Ω resistance

(4) Current in the wire AB is zero.

25) A galvanometer has a resistance of 3663 ohm. A shunt S is connected across it such that (1/34) of the total current passes through the galvanometer. Then, the value of shunt is :-

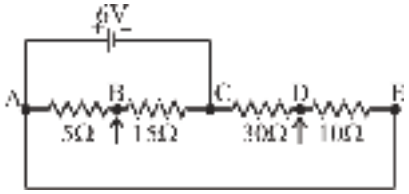
(1) 3663 Ω

(2) 111 Ω

(3) 107.7 Ω

(4) 3555.3 Ω

26) Four resistors are connected as shown in the following figure. A 6 V battery of negligible resistance is connected across terminals A and C. The potential difference across terminals B and D



will be:-

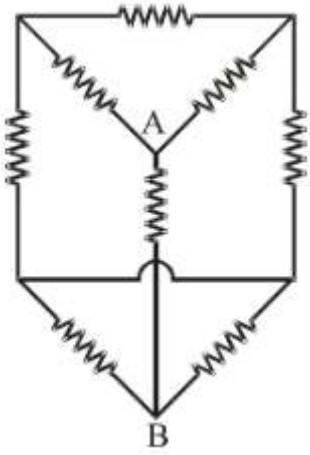
(1) 0 volt

(2) 1.5 volt

(3) 2 volt

(4) 3 volt

27) Effective resistacne between point A and B for following network is-(each branch of resistance is R)



(1) $\frac{5}{3}$ R

(2) $\frac{3}{5}$ R

(3) R

(4) $\frac{R}{2}$

28) Six identical cells each of emf E & internal resistance r, are joined in series to form closed Ckt. Only one cell A is joined with reverse polarity. Then the terminal potential difference across cell (A) is :–

(1) $\frac{E}{3}$

(2) $\frac{5}{3}E$

(3) $\frac{1}{6}E$

(4) $\frac{11}{6}E$

29) Co-efficient of temperature for two wires are (α) and (–3α). Now both the wires are connected in series, if the resistance of their series combination does not depends on temp, then calculate the ratio of their resistances at 0°C:-

(1) 4

(2) 3

(3) 1

(4) 2

30) Three heater wires, made of the same material and having the same length and the same radius, are first connected in series and then in parallel to a constant potential difference. If the rates of heat produced in the two cases are Hs and Hp respectively then Hs/Hp will be :-

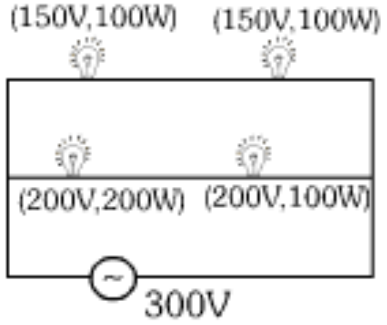
(1) $\frac{1}{3}$

(2) 3

(3) $\frac{1}{9}$

(4) 9

31) In the arrangements of four bulbs as shown in figure total power consumption will be :-



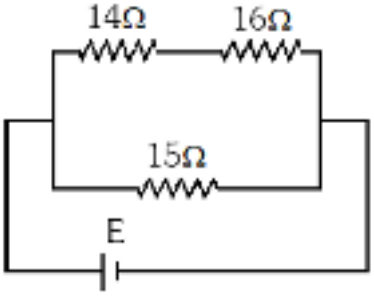
(1) 200 W

(2) 350 W

(3) 500 W

(4) 300 W

32) In the following circuit, the amount of heat emitted by 15Ω resistance is 120 cal/s. The amount of heat emitted by 14 Ω resistance will be (in cal/s) :-



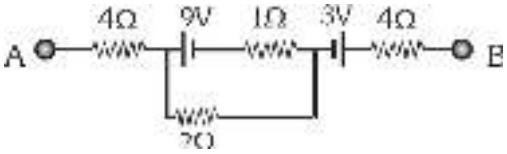
(1) 60

(2) 32

(3) 28

(4) 240

33) In the circuit shown in figure potential difference between point A and B is 16V. Find the current passing through 2Ω resistance.

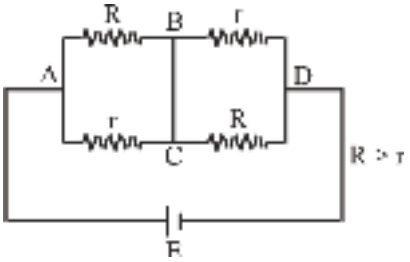


(1) 3.5 A

(2) 0 A

(3) 2.5 A

(4) 4 A



34) For given circuit, select correct alternative

Equivalent resistance between A and D is

(1) $\frac{2Rr}{R+r}$

(2) Current through BC is $\frac{E(R-r)}{2rR}$

(3) Current through BC is from C to B

(4) All of these

35) The time period of a vibration magnetometer is T_0. Its magnet is replaced by another magnet whose moment of inertia is 3 times and magnetic moment is (1/3) of the initial magnet. The time period now will be :

(1) $3T_0$

(2) $T_0/\sqrt{3}$

(3) T_0

(4) $T_0 / 3$

36) Magnitude of magnetic field (in SI units) at the centre of a hexagonal shape coil of side 10 cm, 50 turns and carrying current I (Ampere) in units of $\frac{\mu_0 I}{\pi}$ is :

(1) $250\sqrt{3}$

(2) $5\sqrt{3}$

(3) $500\sqrt{3}$

(4) $50\sqrt{3}$

37) A square loop is carrying a steady current I and the magnitude of its magnetic dipole moment is m. If this square loop is changed to a circular loop and it carries the same current, the magnitude of the magnetic dipole moment of circular loop will be :

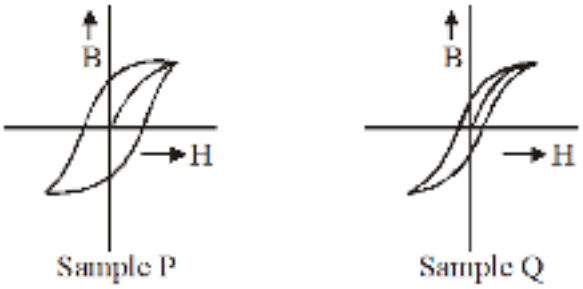
(1) $\frac{3m}{\pi}$

(2) $\frac{4m}{\pi}$

(3) $\frac{2m}{\pi}$

(4) $\frac{m}{\pi}$

38) If the B-H curves of two samples of P and Q of iron are as shown below, then which one of the following statements is **correct** ?



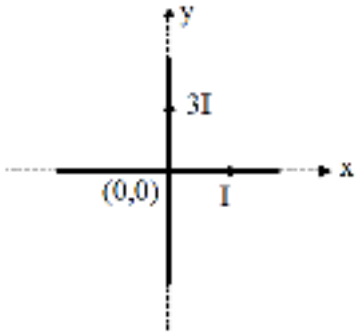
(1) Both P and Q are suitable for making permanent magnet

(2) P is suitable for making permanent magnet and Q for making electromagnet

(3) P is suitable for making electromagnet and Q is suitable for permanent magnet

(4) Both P and Q are suitable for making electromagnets

39) The equation of line on which magnetic field is zero due to system of two perpendicular infinitely long current carrying straight wires, is:



(1) x = y

(2) x = 2y

(3) x = 3y

(4) 3x = y

40) If the acceleration of a charge particle in magnetic field $\vec{B} = 2\hat{i} + a\hat{j} + \hat{k}$ is $-5\hat{i} + 3\hat{j} + a^2\hat{k}$, then the values of a are :-

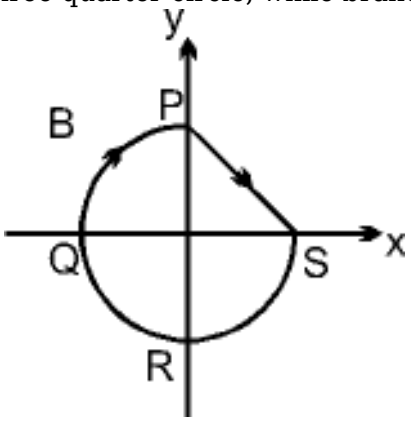
(1) (–2, 5)

(2) (3, 7)

(3) (2, –5)

(4) (–3, 7)

41) A current carrying loop is placed in a uniform magnetic field pointing in negative z direction. Branch PQRS is a three quarter circle, while branch PS is straight. If force on branch PS is F, force on branch PQR is :-



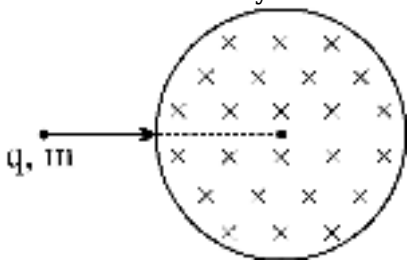
(1) $\sqrt{2}F$

(2) $\frac{F}{\sqrt{2}}$

(3) $\frac{\pi F}{\sqrt{2}}$

(4) $\sqrt{2}\pi F$

42) A particle having mass m, charge q enters a cylinder region having uniform magnetic field B in the inward direction as shown. If the particle is deviated by 60° as it emerges out of the field then what is the time spent by it in the field.



(1) $\frac{2\pi m}{qB}$

(2) $\frac{2\pi m}{3qB}$

(3) $\frac{\pi m}{3qB}$

(4) It depends on the speed of particle.

43) An ionized gas contains both positive and negative ions. If it is subjected simultaneously to an electric field along the –x direction and a magnetic field along the +z direction, then :-

(1) Positve ions deflect +y direction and negative ions towards –y direction

(2) All ions deflect towards +y direction

(3) All ions deflect towards –y direction

(4) Positive ions deflect towards –y direction and negative ions towards +y direction

44) An electron is moving on a circular path of radius r with speed v in a transverse magnetic field B e/m for it will be :-

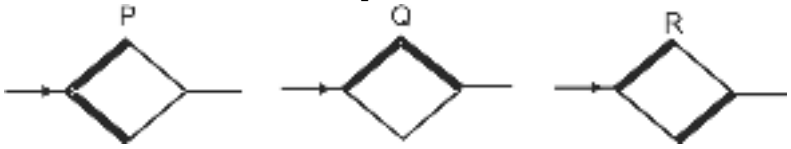
(1) $\frac{v}{Br}$

(2) $\frac{B}{rv}$

(3) Bvr

(4) $\frac{vr}{B}$

45) Two thick wires and two thin wires, all of the same materials and same length form a square in the three different ways P, Q and R as shown in figure with current connection shown. The magnetic field at the centre of the square is zero in cases:-



(1) in P only

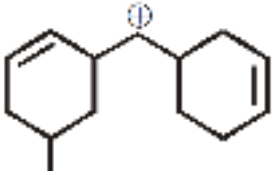
(2) in P and Q only

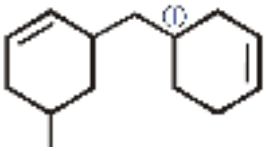
(3) in Q and R only

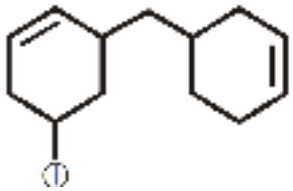
(4) P and R only

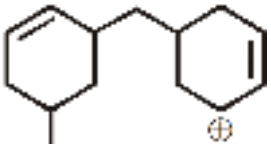
CHEMISTRY

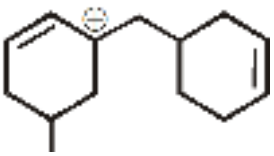
1) Which of the following is the correct stability order for given carbocations ?

(1) 

(2) 

(3) 

(4) 

(5) 

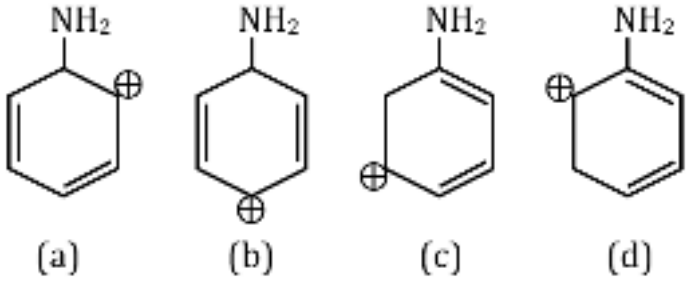
(1) 5 > 4 > 2 > 1 > 3

(2) 3 > 4 > 2 > 1 > 5

(3) 3 > 5 > 2 > 1 > 4

(4) 1 > 3 > 5 > 4 > 2

2) The most stable carbocation for the following is:



(1) c

(2) d

(3) b

(4) a

3) **Statement 1 :**  furan is aromatic compound.

Statement 2 : Furan has cyclic delocalisation of 6 π electron, so it follow Huckel's rule.

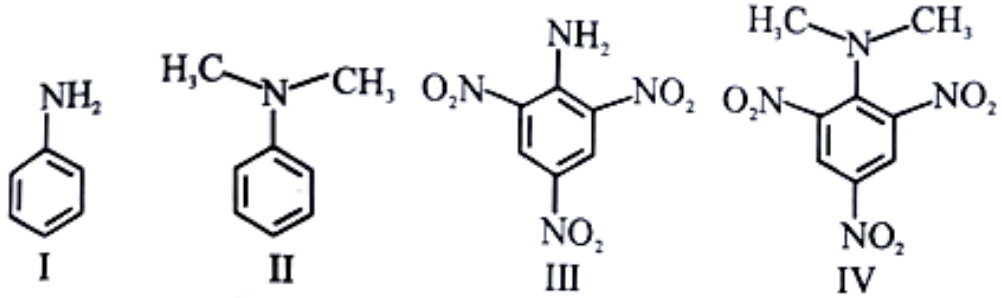
(1) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(2) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(3) Statement-1 is true, statement-2 is false.

(4) Statement-1 is false, statement-2 is true.

4) Consider the following four compounds I, II, III and IV.



Choose the correct statements(s)

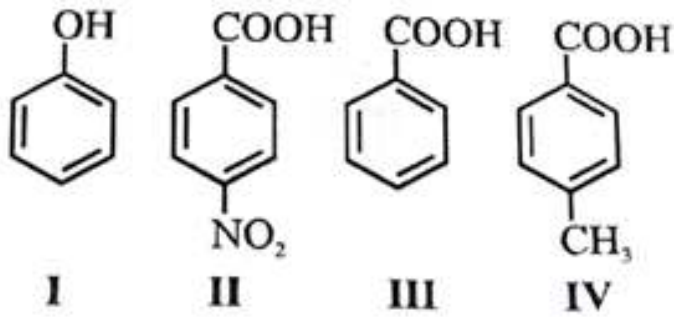
(1) The order of basicity is II > I > III > IV.

(2) I is more basic then II due to less crowding

(3) Resonance effect is less in III than in IV.

(4) Steric effect makes compound IV more basic than III.

5) The correct order of acidic character of the following compounds is:



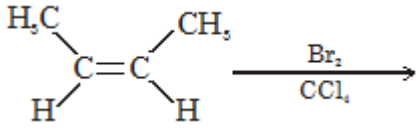
(1) III > II > I > IV

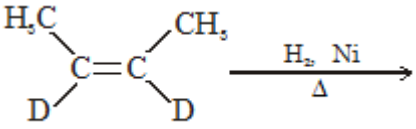
(2) IV > III > II > I

(3) I > II > III > IV

(4) II > III > IV > I

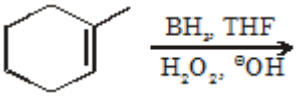
6) How many of the following addition reactions are syn addition reaction.

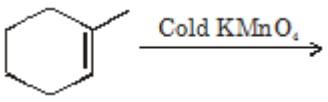
(a) 

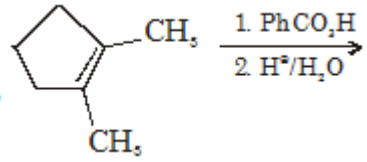
(b) 

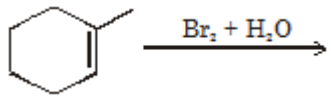
(c) $H_3C-C\equiv C-CH_3 \xrightarrow{H_2,\ Pd\text{–}BaSO_4}$

(d) $H_3C-C\equiv C-CH_3 \xrightarrow{Na,\ Liq.\ NH_3}$

(e) 

(f) 

(g) 

(h) 

(1) 5

(2) 6

(3) 3

(4) 4

7)

NBS react with 1-butene to give

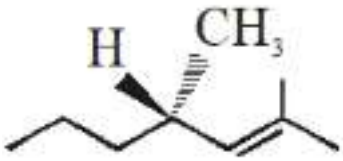
(1) 1,2-dibromobutane

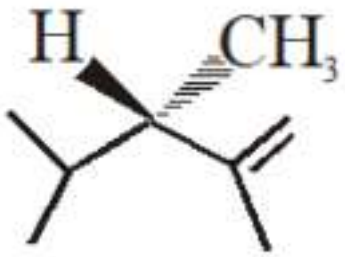
(2) 3-bromobut-1-ene

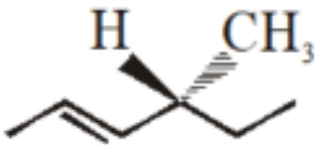
(3) 1-bromobut-1-ene

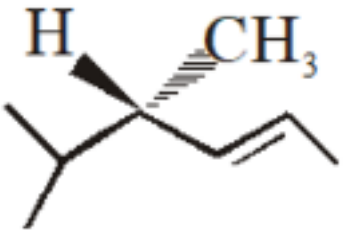
(4) 1,2-dibromobut-1-ene

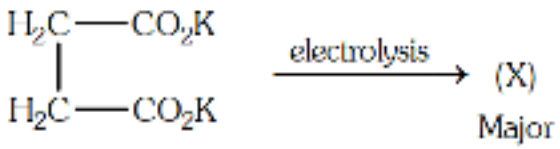
8) Which of the following compounds produces an optically inactive compound on hydrogenation ?

(1) 

(2) 

(3) 

(4) 

9) 

Product (X) of the reaction is :

(1) Ethane

(2) Ethene

(3) Cyclobutane

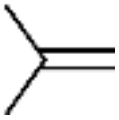
(4) Butene

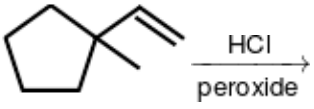
10) Which of the following will react least readily with HBr (electrophilic addition) ?

(1) CH_3–C≡C–CH_3

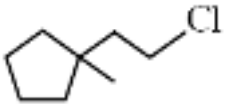
(2) CH_3–CH=CH–CH_3

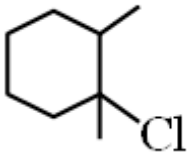
(3) Ph–CH=CH_2

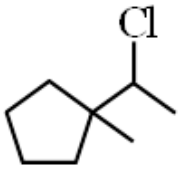
(4) 

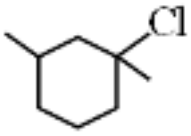
11) 

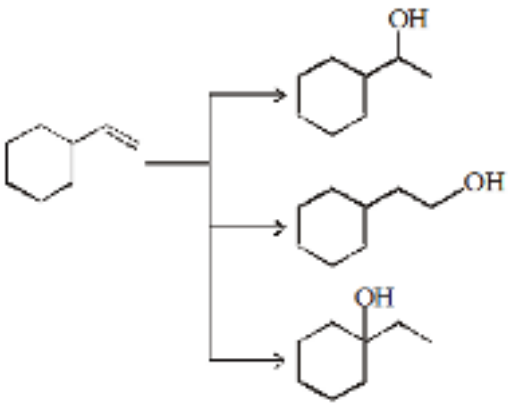
M. M is :-

(1) 

(2) 

(3) 

(4) 

12) 

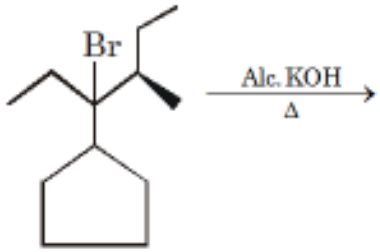
Reactions P, Q and R should be:-

(1) Simple addition of water to alkene.

(2) Hydration of alkene, Hydroboration oxidation and oxymercuration-Demercuration.

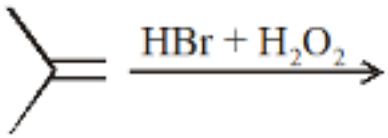
(3) Oxymercuration-Demercuration, Hydroboration-oxidation and hydration of alkene.

(4) Hydroboration-oxidation, oxymercuration-Demercuration and Hydration of alkene.

13) 

Number of possible eliminaton products of the above reaction

(1) 3
(2) 2
(3) 4
(4) 5

14) 

Product., Product is :

(1) 3° alkyl bromide
(2) 1° alkyl bromide
(3) 2° alkyl bromide
(4) 1° alcohol

15) The reactivity of alkyl halides for Wurtz reaction is :

(1) $1^0 > 2^0 > 3^0$
(2) $3^0 > 2^0 > 1^0$
(3) $2^0 > 3^0 > 1^0$
(4) $1^0 > 3^0 > 2^0$

16) Which have linear shape :-

(1) XeF_2
(2) I_3^-
(3) CO_2
(4) All

17) Find the incorrect statement :-

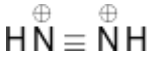
(1) Number of hybrid orbitals produced is equal to the number of atomic orbitals that get hybridised.
(2) Hybrid orbitals are equivalent in energy and shape
(3) Hybrid orbitals are more effective in forming sigma bonds than pure atomic orbitals
(4) In hybridisation only half filled atomic orbitals participates

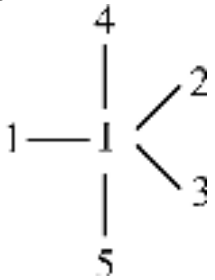
18) The highest amount of s-character is observed in:

(1) N–H bond of NH_3

(2) N–H bond of NH_4^+

(3) N–H bond in $H_2N–NH_2$

(4) N–H bond in 

19) In order that the diagram below correctly depict the structure of ICl_2^- , at what position should the Cl atoms be drawn ? 

(1) 1, 2

(2) 1, 4

(3) 2, 3

(4) 4, 5

20) Out of $N_3^\ominus, I_3^\ominus, I_3^+$ and BrF_3 Linear species are ?

(1) $N_3^\ominus, BrF_3$

(2) $N_3^\ominus, I_3^+$

(3) $N_3^\ominus, I_3^\ominus$

(4) $I_3^\ominus, BrF_3$

21) Match the following :

Bonds		**BE (kJ mole^{-1})**	
(i)	O–O	(p)	467
(ii)	O–H	(q)	146
(iii)	S–S	(r)	374
(iv)	S–H	(s)	266

Correct match is :

(1) i-q, ii-p, iii-s, iv-r

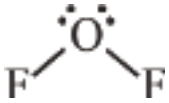
(2) i-q, ii-p, iii-r, iv-s

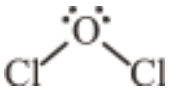
(3) i-p, ii-q, iii-s, iv-r

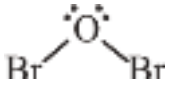
(4) i-p, ii-q, iii-r, iv-s

22) Which has highest bond angle ?

(1) 

(2) 

(3) 

(4) 

23) In which coordinate bond is present ?

(1) NO_2^-

(2) NO_3^-

(3) CO_3^{-2}

(4) All having coordinate bond

24) The formal charges on the three O-atoms in O_3 molecule are :-

(1) 0, 0, 0

(2) 0, 0, –1

(3) 0, 0, +1

(4) 0, + 1,–1

25) Which of the following has been arranged in order of decreasing dipole moment ?

(1) $CH_3Cl > CH_3F > CH_3Br > CH_3I$

(2) $CH_3F > CH_3Cl > CH_3Br > CH_3I$

(3) $CH_3Cl > CH_3Br > CH_3I > CH_3F$

(4) $CH_3F > CH_3Cl > CH_3I > CH_3Br$

26) In O_2^-, O_2 & O_2^{-2} number of antibonding electron's respectively are -

(1) 3, 2, 4

(2) 7, 6, 7

(3) 7, 6, 8

(4) 8, 6, 8

27) Four diatomic species are listed below in different sequences. Which of these presents the correct order of their increasing bond order ?

(1) $C_2^{2-} < He_2^+ < NO < O_2^-$

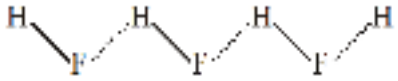
(2) $He_2^+ < O_2^- < NO < C_2^{2-}$

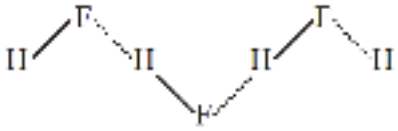
(3) $O_2^- < NO < C_2^2 < He_2^+$

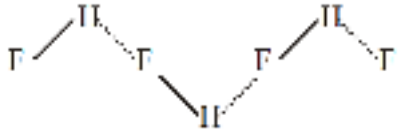
(4) $NO < C_2^{2-} < O_2^- < He_2^+$

28) The H-bond in solid HF can best represented as :-

(1) H – F ..... H – F ... H – F

(2) 

(3) 

(4) 

29) Correct order of ionic character

(1) $AlCl_3 > MgCl_2 > NaCl$

(2) $MgO > NaCl > CaCl_2$

(3) $RbCl > CsCl > NaCl$

(4) $CsCl > RbCl > NaCl$

30) **Assertion :** 2nd period elements do not expand their octet. **Reason :** 2nd period elements do not have vacant 2d-orbitals.

(1) Assertion is true but the reason is false

(2) Assertion is false but reason is true

(3) Both assertion and reason are true and the reason is the correct explanation of assertion

(4) Both assertion and reason are true but reason is not the correct explanation of assertion

31) An aq. solution is prepared by dissolving 0.1 mole H_2CO_3 in sufficient water to get 100 ml solution at 25°C.
$ka_1(H_2CO_3) = 4 \times 10^{-6}$;
$ka_2(H_2CO_3) = 5 \times 10^{-11}$.
The only incorrect equilibrium concentration is -

(1) $[H^+] = 6.32 \times 10^{-4}$ M

(2) $[HCO_3^-] = 2 \times 10^{-3}$ M

(3) $[CO_3^{2-}] = 5 \times 10^{-11}$ M

(4) $[OH^-] = 5 \times 10^{-12}$ M

32) In which of the following combination, is buffer action expected ?
(a) $NH_4OH + NH_4Cl$
(b) HCl + NaCl
(c) NH_4OH + HCl in 2 : 1 mole ratio
Select the correct answer using the code given below :

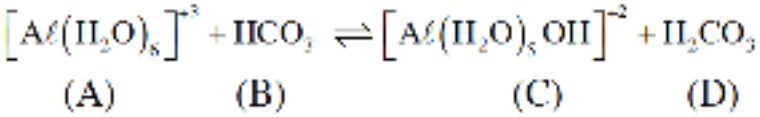
(1) a & b

(2) a & c

(3) a, b & c

(4) None of these

33) In the following :



species behaving as bronsted – lowry acid are

(1) (A), (D)

(2) (A), (C)

(3) (B), (C)

(4) (B), (D)

34) A buffer that is a mixture of acetic acid ($K_a = 2 \times 10^{-5}$) and potassium acetate has pH = 5.18. The $\frac{[CH_3COO^-]}{[CH_3COOH]}$ ratio in this buffer is approx :-

(1) 1 : 1

(2) 3 : 1

(3) 5 : 1

(4) 1 : 3

35) The solubility of $Ba_3(AsO_4)_2$ (molar mass = 690) is 6.9×10^{-2} g/100 mL. What is the value of its K_{sp} ?

(1) 1.08×10^{-11}

(2) 1.08×10^{-13}

(3) 1.0×10^{-15}

(4) 6.0×10^{-13}

36)

The correct order of increasing solubility of AgCl in :
(a) Water (b) 0.1 M NaCl
(c) 0.1 M $BaCl_2$ (d) 0.1 M NH_3 is :

(1) a < b < c < d

(2) b < c < a < d

(3) c < b < d < a

(4) c < b < a < d

37) The number of moles of ferrocyanide ions required to reduce 1 mole of permanganate ion in the following redox reaction. $[Fe(CN)_6]^{4-} + MnO_4^- + H^+ \rightarrow CO_2 + Mn^2 + NO_3^- + Fe^{3+} + H_2O$

(1) $\frac{5}{61}$

(2) $\frac{61}{5}$

(3) $\frac{5}{11}$

(4) $\frac{11}{5}$

38) Which buffer is present in human blood :-

(1) $NaH_2PO_4 + Na_2HPO_4$

(2) $H_3PO_4 + NaH_2PO_4$

(3) $CH_3COOH + CH_3COONa$

(4) $H_2CO_3 + HCO_3^-$

39) $N_{2(g)} + 3H_{2(g)} \rightleftharpoons 2NH_{3(g)}$
Initially the molar ratio of N_2 and H_2 was 1 : 3. At equilibrium 50% of each reactant has reacted. If the equilibrium pressure is P then the partial pressure of NH_3 at equilibrium is :-

(1) $\frac{P}{3}$

(2) $\frac{P}{4}$

(3) $\frac{P}{6}$

(4) $\frac{P}{8}$

40) For a reaction $N_2 + 3H_2 \rightleftharpoons 2NH_3$, the value of K_C does not depends upon :-
(a) Initial concentration of the reactants
(b) Pressure (c) Temperature
(d) Catalyst

(1) Only a

(2) a, b, c

(3) a, b, d

(4) a, b, c, d

41) For the reaction : $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$, the degree of dissociation (α) of HI(g) is related to equilibrium constant K_P by the expression :

(1) $1 + 2\sqrt{K_P}$

(2) $\sqrt{\frac{1 + 2K_P}{2}}$

(3) $\sqrt{\frac{2K_P}{1 + 2K_P}}$

(4) $\frac{2\sqrt{K_P}}{1+2\sqrt{K_P}}$

42)

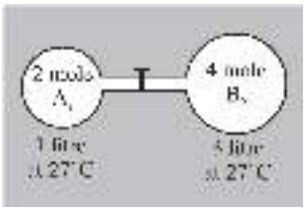
The equilibrium constant for the reaction $H_2(g) + S(s) \rightleftharpoons H_2S(g)$; is 18.5 at 925 K and 9.25 at 1000 K respectively. The enthalpy of the reaction will be :

(1) – 68000.05 J mol^{-1}

(2) –71080.57 J mol^{-1}

(3) – 80071.75 J mol^{-1}

(4) 57080.75 J mol^{-1}

43) 

The gas A_2 in the left flask allowed to react with gas B_2 present in right flask as $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$; Kc = 4 at 27°C. What is the concentration of AB when equilibrium is established?

(1) 1.33 M

(2) 2.66 M

(3) 0.66 M

(4) 0.33 M

44) A compound of Xe and F is found to have 53.5% Xe. What is the oxidation number of Xe in this compound? (At. wt. of Xe = 131):-

(1) –4

(2) +2

(3) +4

(4) +6

45) Nitric oxide acts as a reducing agent in the reaction:

(1) $4NH_3 + 5O_2 \rightarrow 2NO + 6H_2O$

(2) $2NO + 3I_2 + 4H_2O \rightarrow 2NO_3^- + 6I^- + 8H^+$

(3) $2NO + H_2SO_3 \rightarrow N_2O + H_2SO_4$

(4) $2NO + H_2S \rightarrow N_2O + S + H_2O$

BIOLOGY

1) Ground tissue is not differentiated in layers in –

(1) Monocot root

(2) Dicot root

(3) Dicot stem

(4) Monocot stem

2) Find the correct option with respect to vascular tissue of dicotyledonous stem :-

(1) Conjoint, closed, exarch, collateral

(2) Radial, endarch, open, bicollateral

(3) Radial, endarch, closed, collateral

(4) Conjoint, collateral, open, endarch

3) Read the different components from (a) to (d) in the list given below and find the correct order of the components with reference to their arrangement from outer side to inner side in a woody dicot stem:
(a) Secondary cortex
(b) Wood
(c) Secondary phloem
(d) Phellem
The **correct** order is :

(1) (d), (c), (a), (b)

(2) (c), (d), (b), (a)

(3) (a), (b), (d), (c)

(4) (d), (a), (c), (b)

4) Which of the following statement is **true** for heart wood ?

(1) Conducts water and provides mechanical support

(2) Light coloured and provides mechanical support

(3) Neither conducts water nor provides mechanical support

(4) Does not conduct water but provides mechanical support

5) In dorniventral leaf palisade parenchyma is present at

(1) Adaxial surface

(2) Abaxial surface

(3) Both Adaxial and Abaxial

(4) Sometimes adaxial sometimes abaxial

6) Choose the correct and **incorrect** statements :-
(a) All the tissues on the inner side of the endodermis forms stele.
(b) Pith is large and well developed in dicot root
(c) Conjuctive tissue present in both stem and root
(d) In monocot stem, water containing cavities are present within the vascular bundles
(e) Bulliform cells occurs in grasses.

(1) a, b, e-correct and c, d-incorrect

(2) a, b, d, e-correct and c-incorrect

(3) a, d, e-correct, and b, c-incorrect

(4) a, e-correct, and b, c, d-incorrect

7) Suberin in chiefly deposited in the cells of :

(1) Sclerenchyma

(2) Collenchyma

(3) Cork

(4) Phelloderm

8) In which of the following collateral vascular bundles with endarch xylem are found ?
(a) Young root of Bean
(b) Maize root
(c) Young stem of Bean
(d) Maize stem

(1) (a) and (b) only

(2) (c) and (d) only

(3) (a) and (c) only

(4) (b) and (d) only

9) Bulliform (Motor) cells are found in

(1) Lower epidermis of monocot leaf

(2) Upper epidermis of monocot leaf

(3) Lower epidermis of dicot leaf

(4) Upper epidermis of dicot leaf

10) Sclerenchymatous pericycle is located just above phloem bundles in patches in :-

(1) Dorsiventral leaf

(2) Monocotyledonae root

(3) Maize stem

(4) Sunflower stem

11) Root apical meristem occupies the_____of a root while the shoot apical meristem occupies the_____ most region of the stem axis :

(1) Base, Proximal

(2) Base, Distant

(3) tip, Distant

(4) tip, Proximal

12)

Which statement is not **correct** :

(1) In grasses, the guard cells are dumb-bellshaped

(2) Subsidiary cells are not a part of stomatal appratus

(3) On the stem the epidermal hairs are called trichomes

(4) The root hairs are unicellular elongations of the epidermal cell

13)

Match the column :-

Column I		**Column II**	
(a)	Jan Ingenhousz	(i)	First action spectrum of photosynthesis
(b)	F.F. Blackman	(ii)	Water is source of O_2 in photosynthetic reaction in green plants
(c)	Joseph Priestley	(iii)	Law of limiting factor
(d)	Julius Von sachs	(iv)	Essentiality of sunlight to plants
(e)	T.W. Engelmann	(v)	Essential role of air in the growth of green plants

(1) a - iv, c - v, d - i

(2) a - iv, c - v, e - i

(3) b - iii, d - v, e - i

(4) c - iv, d - v, e - ii

14) Photosynthesis is :-

(1) Redox-reaction

(2) Endergonic

(3) Anabolic

(4) All of the above

15) Which scientist studied purple and green bacteria to demonstrate that photosynthesis is essentially a light dependent reaction in which hydrogen from a suitable oxidisable compound reduces carbon dioxide to carbohydrate ?

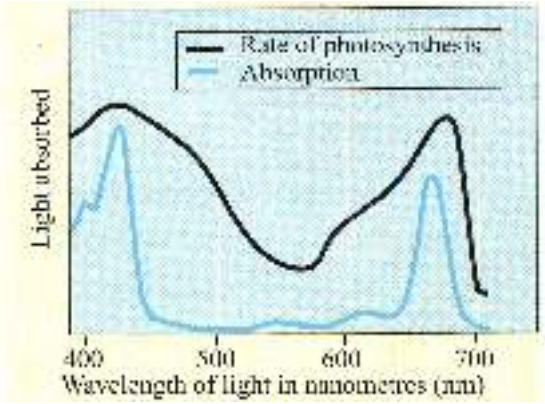
(1) T.W. Engelmann

(2) Julius Von Suchs

(3) Cornelius Van Niel

(4) Jan Ingenhousz

16) Represented below is the graph showing action spectrum of photosynthesis superimposed on absorption spectrum of chlorophyll 'a'. Which one of the following is correct for this ?



(1) There is a complete one to one overlap between the absorption and action spectrum

(2) By this overlapping it can be concluded that chlorophyll 'a' is the chief pigment of photosynthesis

(3) This overlapping shows that only chlorophyll 'a' absorb the entire light energy

(4) These graphs show that photosynthesis occurs only in the wavelengths of blue and red light

17) Find the colour at which chlorophyll-a shows the maximum absorption :-

(1) Blue

(2) Red

(3) Green

(4) Violet

18) **Statement I :** CO_2 is the major limiting factor for photosynthesis.
Statement II : C_4 plants adapted to dry tropical regions

(1) Only statement I is correct

(2) Only statement II is correct

(3) Both statements I and II are incorrect

(4) Both statements I and II are correct

19) Number of light quanta needed in yield of one O_2 is-

(1) Quantum yield

(2) Quantum requirement

(3) Quantum conversion

(4) Quantasome

20)

Mesophyll cell, Chloroplast, Light, Atmospheric CO_2 concentration, temperature, Water, Amount of chlorophyll.
How many of the above belongs to external factors which affects photosynthesis?

(1) 5

(2) 8

(3) 4

(4) 6

21)

Which of the following is the most crucial step of the Calvin cycle ?

(1) Triose phosphate → Sucrose

(2) 3 - phosphoglycerate → Triose phosphate

(3) Ribulose-1, 5-bisphosphate + CO_2 → 3-phosphoglycerate

(4) Triose phosphate → starch

22)

Saturation of CO_2 concentration for maize and wheat respectively are :-

(1) 450 μL^{-1} and 360 μL^{-1}

(2) 360 μL^{-1} and 450 μL^{-1}

(3) 360 μL^{-1} and 360 μL^{-1}

(4) 450 μL^{-1} and 450 μL^{-1}

23) In photorespiration -

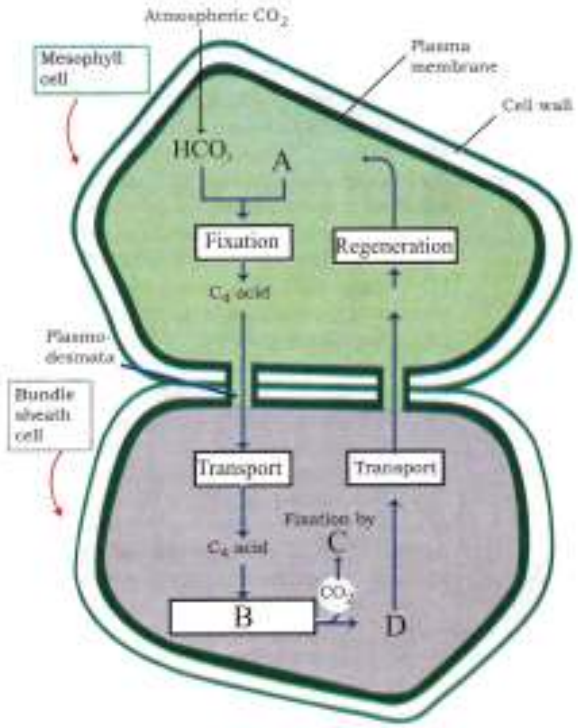
(1) Neither sugar nor ATP is synthesised

(2) Sugar is synthesised

(3) ATP is synthesised but not sugar

(4) Both ATP and sugar are synthesised

24) In the given figure identify A,B,C and D :-



	A	B	C	D
1	PEP	Decarboxylation	C_4 Cycle	Pyruvate
2	Pyruvate	Decarboxylation	Calvin cycle	C_3 acid
3	Pyruvate	Carboxylation	C_4 Cycle	C_3 acid
4	PEP	Decarboxylation	Calvin cycle	C_3 acid

(1) 1

(2) 2

(3) 3

(4) 4

25) Find the **incorrect** statement :-

(1) In bundle sheath cells of C_4 plants, first product of CO_2 fixation is 3C compound.

(2) In photorespiration both chloroplast and peroxisome are involved

(3) PEP carboxylase plays important role in synthesis of sugar in mesophyll cells of C_4 plants.

(4) ATP requirement of C_4 plants is higher as compared to C_3 plants to produce one molecule of sugar

26) C_4 plants like sugarcane use C_3 cycle to synthesize sugar then what is the significance of operation of C_4 cycle in these plants ?

(1) Kranz anatomy helps them to endure high light intensity.

(2) Lack of photorespiration helps them maintaining low O_2 concentration in Bundle sheath cell

(3) Eftective pumping of CO_2 near site of RuBisCO activity helps maintain high rate of photosynthesis

(4) Low temperature sensitive enzymes helps to adapt to tropical conditions.

27) Which of the following components are required for chemiosmosis ?

(1) A membrane, a proton pump, an electron gradient and ATPase
(2) F_0–F_1 particle, ATP, a membrane and proton gradient
(3) A proton pump, a membrane, a proton gradient and ATPase
(4) A proton pump, a membrane, a proton gradient and Light

28) During Z-scheme of photosynthesis which of the following not helps in creation of proton gradient?

(1) Release of protons in stroma by water splitting
(2) Presence of H-carriers in the thylakoid membrane
(3) Reduction of $NADP^+$
(4) Oxidation of water by photosystem-II

29) The thylakoid membrane bears several F_0 – F_1 particle (ATPase/ATP synthase). Which of following is incorrect for these particles?

(1) One of its part (F_0) is embedded in the membrane and forms transmembrane channel that carries out facilitated diffusion of protons across the membrane.
(2) Its other part (F_1) protrudes out facing towards the stroma
(3) The catalytic sites for ATP formation are located in F_1 part
(4) F_1 is intrinsic protein

30) **Assertion :** In C_4 plants photorespiration does not occur.
Reason : They have a mechanism that increases the concentration of CO_2 at the enzyme site.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
(3) Assertion is True but the Reason is False.
(4) Both Assertion & Reason are False.

31) **Statement-I :** The binding of CO_2 or O_2 to RuBisCO enzyme is competitive, relative concentration of O_2 and CO_2 determines which of the two will bind to the enzyme.
Statement-II : In C_4 plants photorespiration does not occur because they have a mechanism that increases the concentration of CO_2 at the RuBisCO enzyme site.
Regarding the above statements, choose the correct answer from the options given below :-

(1) Both statement-I and statement-II are incorrect
(2) Both statement-I and statement-II are correct
(3) Statement-I is correct but statement-II is incorrect
(4) Statement-I is incorrect but statement-II is correct

32) What is the **correct** sequence of cell organelles during photorespiration?

(1) Chloroplast-Golgi bodies-mitochondria

(2) Chloroplast-Rough Endoplasmic reticulum, Dictyosome

(3) Chloroplast-peroxisome-mitochondria

(4) Chloroplast-vacuole-peroxisome

33) In CAM plants CO_2 acceptor during day time is-

(1) RUBP

(2) PEP

(3) OAA

(4) PGA

34) The biosynthetic phase of photosynthesis
(A) Involves use of ATP & NADPH(H^+)
(B) Takes place in the fluid stroma of chloroplast
(C) Does not depend directly in the presence of light
The **correct** statement are:

(1) only B & C

(2) only A & B

(3) A, B & C

(4) only A & C

35) The final electron acceptor in non-cyclic photophosphorylation is

(1) H_2O

(2) ADP

(3) $NADP^+$

(4) NAD^+

36) **Statement-I :** Mammals and marine fishes mainly excrete urea.
Statement-II : Reptiles and land snails excrete uric acid.

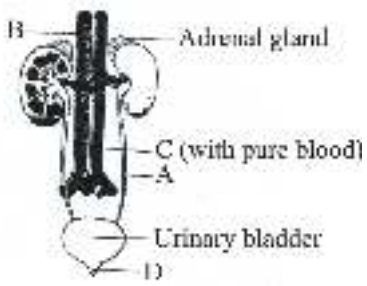
(1) Statement-I and statement-II are correct

(2) Statement-I and statement-II are incorrect

(3) Statement-I is correct and statement-II is incorrect

(4) Statement-I is incorrect and statement-II is correct

37) Observe the following figure ?



Identify A to D structure -

	A	B	C	D
1.	Superior vena cava	Inferior vena cava	Dorsal Aorta	Urethra
2.	Inferior vena cava	Superior vena cava	Dorsal Aorta	Urethra
3.	Ureter	Inferior vena cava	Dorsal Aorta	Urethra
4.	Dorsal Aorta	Inferior vena cava	Urethra	Ureter

(1) 1

(2) 2

(3) 3

(4) 4

38) Match the excretory functions of section I with the parts of the excretory system in section II. Choose the correct combinations from among the answer given-

Section–I		Section–II	
(i)	Ultrafiltration	(a)	Henle's loop
(ii)	Concentration of urine	(b)	Ureter
(iii)	Transport of urine	(c)	Urinary bladder
(iv)	Storage of urine	(d)	Malpighian corpuscles
		(e)	Proximal convoluted tubules

(1) (i) – (d), (ii) – (a), (iii) – (b), (iv) – (c)

(2) (i) – (d), (ii) – (c), (iii) – (b), (iv) – (a)

(3) (i) – (e), (ii) – (d), (iii) – (a), (iv) – (c)

(4) (i) – (e), (ii) – (d), (iii) – (a), (iv) – (b)

39) Human kidney can produce urine nearly four times concentrated than the ..... ?

(1) filtrate present in glomerulus

(2) Filtrate present in bowmans capsule

(3) filtrate present in loop of Henle

(4) filtrate present in afferent arteriole

40) **Statement-I :** The descending limb continues as another highly coiled tubule called distal convoluted tubule (DCT).
Statement-II : The PCT of single nephron open into a straight tube called collecting duct, many of which converge and open into the renal pelvis through medullary pyramids in the calyces.

(1) Statement I is true and statement II is false.

(2) Statement I is false and statement II is true.

(3) Both statement I and II are true.

(4) Both statement I and II are false.

41) Select the incorrect match refered as material as absorbed in different parts of the nephron of kidney ?

	Part of nephrone	**Reabsorbed material**
(1)	Ascending limb of LOH	NaCl
(2)	Descending limb of LOH	Water
(3)	PCT	Glucose
(4)	DCT	Amino acid

(1) 1

(2) 2

(3) 3

(4) 4

42) Correctly match statement are :-

(1) DCT → Formed by continuation of ascending limbs.

(2) Henle's loop → Found maximum in cortex region.

(3) Urine formation → Mainly by reabsorption and zero secretion.

(4) Juxta medullary nephron → Loop of Henle is too short and only extended very little into the medulla.

43) In Hemodialysis cleared blood from machine is pumped back to the body through ?

(1) Artery

(2) Vein

(3) Capillary

(4) Cellophane tubes

44) Counter current mechanism is mainly caused up by

(1) NaCl and Urea

(2) Urea and NH_3

(3) NaCl and other electrolytes

(4) Urea and water

45) NaCl is transported by ascending limb of _____which is exchanged with the descending limb of_____

(1) henle's loop, vasa Recta

(2) Vasa Recta, henle's loop

(3) Medullary interstitium, Vasa Recta

(4) Vasa Recta, collecting duct

46) Given below are two statemetns one is labelled as Assertion (A) and other is labelled as Reason ?
Assertion (A) :- The henle's loop and vasa Recta play a significant Role in counter current
Reason (R) :- Flow of filtration in the two limb's of henle's loop in opposite directions and thus forms a counter current ?
In the light of above statements, choose the current answer from the options given below.

(1) Assertion is true but Reason is false

(2) Assertion is false but Reason is true

(3) Both Assertion and Reason are true and Reason is the correct explanation of (A)

(4) Both Assertion and Reason are true and Reason is not the correct explanation of Assertion

47) In the following how many structure are situated in the cortical region of kidney ?
Malpighian corpuscle, PCT, DCT, Loop of hencle Renal pelvis, vasa recta.

(1) 2

(2) 3

(3) 4

(4) 5

48) Identify the **incorrect** statement from the following

(1) An increase in blood flow to the atria of Heart can cause the Release of Atrial Natriuretic factor (ANF)

(2) ANF can cause vasodilation and thereby decrease the blood pressure

(3) ANF mechanism, Act as check on the renin - angiotensin mechanism.

(4) A decrease in blood pressure can increase the glomerular blood flow and thereby the GFR.

49) An excessive loss of fluid from the body can activate these receptors which stimulate the_____to release _____from the neurohypophysis.

(1) Hypothalamus, Anti diuretic hormone

(2) JGA, Angiotensin II

(3) Hypothalamus, Atrial Natriuretic factor

(4) Hypothalamus, Angiotensin-II

50) Identify the **correct** statement in the following -
(i) The human skin have sweat and sebaceous gland which eliminate certain substance
(ii) Sebum is a watery fluid containing NaCl, Small amount of urea, Lactic acid.
(iii) Primary function of sweat is to faciliate Removal of waste products.

(1) Only i

(2) Only ii

(3) Only iii

(4) Only ii,iii

51) Which of the following is a **correct** statement ?

(1) Z-lines anchor myosin (thick) filament to the ends of the sarcomere.
(2) Sarcoplasmic reticulum stores acetylcholine.
(3) Myosin cross bridges contain calcium binding sites.
(4) Actin and regulatory proteins are located in thin filament.

52) Match **List - I** with **List - II**.

List-I		**List-II**	
(a)	Scapula	(i)	Cartilaginous joints
(b)	Cranium	(ii)	Flat bone
(c)	Sternum	(iii)	Sutures
(d)	Vertebral column	(iv)	Triangular flat bone

Choose the **correct** answer from the options given below.

	(a)	**(b)**	**(c)**	**(d)**
(1)	(i)	(iii)	(ii)	(iv)
(2)	(ii)	(iii)	(iv)	(i)
(3)	(iv)	(ii)	(iii)	(i)
(4)	(iv)	(iii)	(ii)	(i)

(1) 1
(2) 2
(3) 3
(4) 4

53) During muscular contraction which of the following events occur ?
(a) 'H' zone disappears
(b) 'A' band widens
(c) 'I' band reduces in width
(d) Myosin hydrolyzes ATP, releasing the ADP and Pi
(e) Z-lines attached to actins are pulled inwards
Choose the **correct** answer from the options given below.

(1) (a), (c), (d), (e) only
(2) (a), (b), (c), (d) only
(3) (b), (c), (d), (e) only
(4) (b), (d), (e), (a) only

54) Match the columns I and II, and choose the correct combination from the options given.

Column-I		Column-II	
a.	Ball and socket joint	i.	Between carpal and metacarpal of thumb
b.	Hinge joint	ii.	Between humerus and pectoral girdle
c.	Pivot joint	iii.	Between atlas and axis
d.	Saddle joint	iv.	Knee joint

(1) a–iv, b–i, c–ii, d–iii

(2) a–ii, b–iv, c–iii, d–i

(3) a–iii, b–iv, c–i, d–ii

(4) a–iii, b–i, c–ii, d–iv

55) Which of the following statements are **correctly** described :-
(a) M-line- Membrane which held thick filament
(b) Z-line- Fibre that bisect 'A' band
(c) H-zone- Central part of thin filament
(d) Myofibril- Striated appearance due to 'A' and 'I' bands.

(1) a, b, c

(2) b, c

(3) b, c, d

(4) a & d

56) Match the following columns and select the correct option

Column - I		Column - II	
(a)	Gout	(i)	Decreased levels of estrogen
(b)	Osteoporosis	(ii)	Low Ca^{++} ions in the blood
(c)	Tetany	(iii)	Accumulation of uric acid crystals
(d)	Muscular dystrophy	(iv)	Auto immune disorder
		(v)	Genetic disorder

(1) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(2) (a)-(iii), (b)-(i), (c)-(ii), (d)-(v)

(3) (a)-(iv), .(b)-(v), (c)-(i), (d)-(ii)

(4) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

57) Select the **correct** option .

(1) 8th, 9th and 10th pairs of ribs articulate directly with the sternum.

(2) 11th and 12th pairs of ribs are connected to the sternum with the help of hyaline cartilage.

(3) Each rib is a flat thin bone and all the ribs are connected dorsally to the thoracic vertebrae and ventrally to the sternum.

(4) There are seven pairs of vertebrosternal, three pairs of vertebrochondral and two pairs of vertebral ribs.

58) Who forms bridge at the open, activated site of actin filament?

(1) Troponin

(2) Tropomyosin

(3) Head of myosin

(4) Tail of myosin

59) Fill in the blanks from the options given below :-
Cilia in Paramecium helps in the ……A…… of food through cytophorynyx and in …..B…… as well.

	A	B
(1)	Movement	Locomotion
(2)	Locomotion	Movement
(3)	Locomotion	Locomotion
(4)	Movement	Movement

(1) 1

(2) 2

(3) 3

(4) 4

60) Fill in the blanks from the options given below :-
All ….A… are movements but all …..B….. are not …….C……. .

(1) Movement, Locomotion, Locomotion

(2) Locomotion, movement, Locomotion

(3) Locomotions, Locomotion, movement

(4) Movement, Movement, Movement

61) The somatic neural system of the peripheral nervous system

(1) Relays impulses from CNS to smooth muscles of body.

(2) Relays impulses from organs to the CNS.

(3) Regulates the movement of substances in hollow visceral organs of body.

(4) Relays impulses from CNS to skeletal muscles.

62) Find out the **true** and **false** statements.
(A) Human skeleton has 206 bones
(B) There are 12 pairs of ribs in human beings
(C) Sternum is present on the dorsal side of the body
(D) All mammals (except of few) have 8 cervical vertebrae
(E) Ribs are bicephalic

(1) ABC—True, DE—False
(2) ABD—True, CE—False
(3) ABE—True, CD—False
(4) ACE—True, BD—False

63) Contraction of muscle fibre takes place by the sliding of :-

(1) Thin filaments over the actin filaments
(2) Thick filaments over the thin filaments
(3) Thin filaments over the thick filaments
(4) Thick filaments over the myosin filaments

64) The skull region articulates with the superior region of the vertebral column, with the help of :-

(1) Odontoid process
(2) Acromian process
(3) Occipital condyles
(4) Trochanter

65)

A canal that passes through mid brain is :

(1) Canal of schlemn
(2) Eustachian canal
(3) Cerebral aqueduct
(4) Semi-circular canal

66) Which of the following part of brain has very convoluted surface in order to provide additional space for many more neurons ?

(1) mid brain
(2) Pons
(3) medulla oblongata
(4) cerebellum

67) Brain stem is made up of -

(1) Mid brain and cerebrum
(2) Mid brain, pons and spinal cord.
(3) Pons, medulla oblongata and spinal cord.
(4) Mid brain, pons and medulla oblongata.

68) Which part of cerebrum converts short term memory into long term memory?

(1) Prefrontal cortex

(2) Limbic lobe
(3) Temporal lobe
(4) Hippocampus

69) The cerebrum wraps around a structure called thalamus, which is :-

(1) a major coordinating centre for sensory and motor signaling
(2) a major coordinating centre for sensory signal only
(3) a major centre for motor signaling
(4) not a nervous part of a brain

70) The **correct** sequence of meninges of brain from inside to outside in human is :

(1) Duramater, Arachnoid, Piameter
(2) Arachnoid, Piameter, Duramater
(3) Piameter, Duramater, Archnoid
(4) Piameter, Arachnoid, Duramater

71) Centres for controlling body temperature, hunger and thirst are located in ?

(1) Hypothalamus
(2) Thalamus
(3) Hippocampus
(4) Amuygdala

72) Sexual behaviour, expression fo emotional reactions etc. are controlled by ?

(1) Limbic system and Epithalamus
(2) Thalamus and Hypothalamus
(3) Corpus Callosum and Hypothalamus
(4) Limbic system and Hypothalamus

73) Mid brain is located between _____(a)_____ of the forebrain and _____(b)_____ of the Hind brain. Identify option which correctly identifies (a) & (b) :

(1) (a) Epithalamus, (b) Pons
(2) (a) Hypothalamus, (b) Cerebellum
(3) (a) Thalamus, (b) Medulla
(4) (a) Hypothalamus, (b) Pons

74) Injury localised to the medulla oblongata would most likely disrupt

(1) Short-term memory
(2) Expression of emotional reactions
(3) Coordination during locomotion
(4) Respiratory rhythm

75) Read the following flow regarding to voltage :-

$-70\ mv \rightarrow -55mv \rightarrow 0mv \rightarrow +10\ mv$

(A) (B) (C) (D) at which potential Na^+–K^+ pump is closed :-

(1) A

(2) A & B

(3) B,C & D

(4) A & D

76) Given below are two statements :
Statement - I : Nissl's granules are found only in the cell body of the neurons.
Statement - II : The PNS comprises of all the nerves of the body associated with the brain and spinal cord.
Choose the correct answer from the options given below:-

(1) Both statements are incorrect.

(2) Statement - I is correct and statement - II is incorrect.

(3) Statement - I is incorrect and statement - II is correct.

(4) Both statements are correct.

77) Select the **incorrect** statement.

(1) Myelinated nerve fibres are enclosed by Schwann cells

(2) Unmyelinated nerve fibres are not enclosed by a Schwann cell in PNS

(3) Unmyelinated nerve fibres are commonly found in autonomous and somatic neural systems

(4) Myelinated nerve fibres are found in spinal and cranial nerves

78) Select the **incorrect** statement :-

(1) Chemicals called neurotransmitters are found in synaptic vesicles.

(2) Terminal end of dendron contain bulb like structure called synaptic knob.

(3) ANS transmits impulses from CNS to the involuntary organs.

(4) Unipolar nervous are usually found in embryonic stage.

79) Depolarisation of the axonal membrane results in development of-

(1) Negative charge on the outer side and positive charge on the inner side

(2) Positive charge on the outer side and negative charge on the inner side

(3) Negative charge on both side

(4) Positive charge on both side.

80) **Assertion :** Cerebral cortex contain large area responsible for intesensory associations, memory and communication.
Reason : Association areas are large region that are neither clearly sensory nor motor in function.

(1) Both assertion & reason are true but reason is not a correct explanation of the assertion.

(2) Both assertion & reason are true & the reason is a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Both assertion & reason are false.

81) Which one of the following statements is not **correct**?

(1) For plants in shade 'or' in dense forest, light act as a limiting factor

(2) As compared to light reactions, dark reactions of photosynthesis are more temperature sensitive

(3) Water stress in plants can reduce the CO_2 availability for photosynthesis

(4) As compared to C_3 plants, C_4 plants respond more to higher CO_2 concentration

82) Vascular bundle of monocot stem is surrounded by-

(1) Sclerenchymatous bundle cap.

(2) Sclerenchymatous bundle sheath

(3) Parenchymatous bundle cap

(4) Parenchymatous bundle sheath

83) Crops such as tomatoes and bell pepper, allowed to grow in a carbon dioxide rich environment, show higher yields because :

(1) They show an increased rate of photosynthesis at higher carbon dioxide concentrations

(2) They can respond to high carbon dioxide conditions even in low light conditions

(3) They show C4 pathway for carbon fixation at high carbon dioxide is the limiting factor in such plants

(4) Only carbon dioxide is the limiting factor in such plants

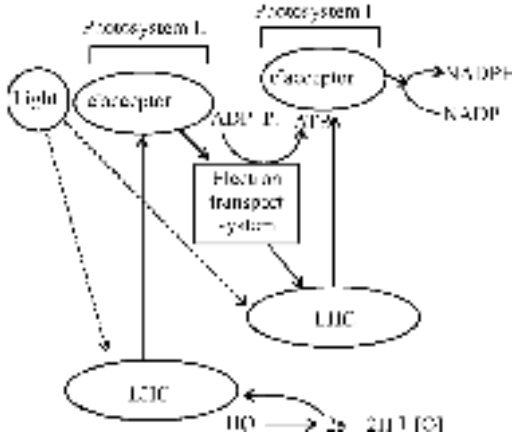
84) Adaxial, vertically elongated, parallelly present parenchymatous tissue that help in photosynthesis found in-

(1) Isobilateral leaf

(2) Dorsi-ventral leaf

(3) Dicot stem

(4) Aquatic plant leaf only

85) 

(A) ATP synthesis can be performed by PS-II alone
(B) ATP synthesis occurs only in PS-I
(C) There is pumping of H^+ in ETC through F_0–F_1
(D) There is pumping of H^+ from stroma to lumen via carrier of ETC
How many statements are **incorrect** ?

(1) 1
(2) 2
(3) 3
(4) 4

86)

The electron flow from PSII to NADP via ETS and PSI follow :

(1) Uphill → Downhill → Uphill → Downhill
(2) Downhill → Uphill → Downhill → Uphill
(3) Downhill → Uphill → Plateau → Uphill
(4) Uphill → Downhill → Plateau → Uphill

87) During light reaction, as electrons move through photosystems, protons are transported across the membrane. This happens because of

(1) The primary acceptor of electron (Located towards the outer surface of the membrane) transfers its electron not to an electron carrier but to H carrier.
(2) The primary acceptor of electron transfer only its electrons to electron carrier
(3) The primary acceptor of electron transfer only H^+ to the next carrier
(4) NADP-reductase is present in grana.

88) Which of the following pair of micronutrients would help in the light phase of photosynthesis to help in the reaction leading to oxygen evolution ?

(1) Manganese and Molybdenum
(2) Molybdenum and Iron
(3) Manganese and Chlorine
(4) Zinc and Chlorine

89) Consider the below equation of light reaction
$NADP^+ + 2H^+ + 2e^- \longrightarrow NADPH + H^+$
Protons for this equation are taken from :-

(1) Stroma of chloroplast
(2) Lumen of thylakoid
(3) Matrix of mitochondria
(4) Cytoplasm

90) Exarch xylem, 3 xylem bundles, and formation of periderm can be seen in-

(1) Dicot stem
(2) Monocot stem
(3) Dicot root
(4) Monocot root

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	4	1	1	4	1	3	1	2	2	4	4	4	1	4	1	1	1	3	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	3	4	4	2	1	2	2	2	3	4	3	1	4	1	3	2	2	3	3
Q.	41	42	43	44	45															
A.	1	3	2	1	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	1	1	4	4	4	2	2	2	1	2	3	4	2	1	4	4	4	4	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	2	4	1	3	2	3	4	3	1	2	1	2	2	4	1	4	1	3
Q.	86	87	88	89	90															
A.	1	2	3	4	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	4	4	4	1	3	3	2	2	4	3	2	2	4	3	2	1	4	2	3
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	2	1	4	3	3	3	1	4	1	2	3	1	3	3	1	3	1	2	4
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	1	2	1	1	3	2	4	1	1	4	4	1	2	4	2	4	3	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	3	3	3	3	4	4	4	1	4	1	4	4	4	3	3	2	2	1	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	2	1	2	3	1	1	3	1	3										

SOLUTIONS

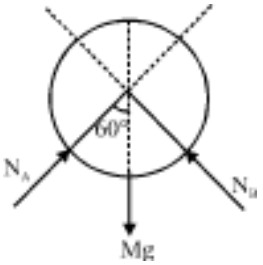
PHYSICS

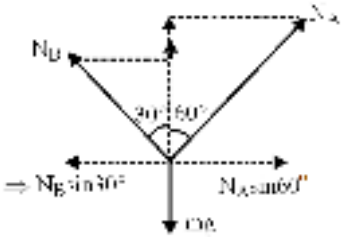
1) **Conceptual**

2) **Conceptual**

3) Acceleration of system = $\frac{F}{5m} = 2\ m/sec^2$
Normal force between B and C is = 3 × 2 N = 6 N
Normal force between A and B is = 4 × 2 = 8 N
Normal force between C and D is = 2 × 2 = 4 N

4) **Conceptual**





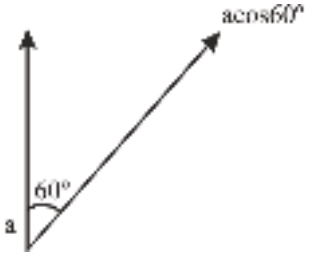
5)
$N_A \sin 60° = N_B \sin 30°$
$N_A \cdot \frac{\sqrt{3}}{2} = N_B \frac{1}{2}$
$\sqrt{3}N_A = N_B$
$\boxed{\sqrt{3}N_A = N_B}$

6)

$a = \frac{f}{m}$
max. $a = \frac{f_{lim}}{m} = \frac{\mu mg}{m}$
$= 0.3 \times 10 = 3\ m/s^2$

7) Initially particle was at rest. By the application of force its momentum increases.
Final momentum of the particle Area of F - t graph ⇒ mu = Area of semi circle
$\Rightarrow mu = \frac{\pi r^2}{2} = \frac{\pi r_1 r_2}{2} = \frac{\pi (F_0)(T/2)}{2} \Rightarrow u = \frac{\pi F_0 T}{4m}$

8) N – mg cos60° = ma cos60°
N = m cos60° (a + g) = 400 N





Reading will be 40 kg-wt

9) F = T

$a = \frac{F}{6m} = \frac{T}{6m}$

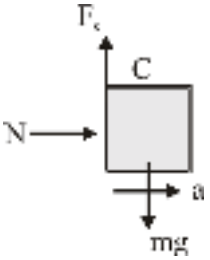
$T_Q = 5m \times a$

$T_Q = 5m \frac{T}{6m}$

10)

Horizontal acceleration of the system is,

$$a = \frac{F}{2m + m + 2m} = \frac{F}{5m}$$



Let N be the normal reaction between B and C.

Free body diagram of C gives

$$N = 2ma = \frac{2}{5}F$$

Now, B will not slide downward if

$\mu N \geq m_B g$

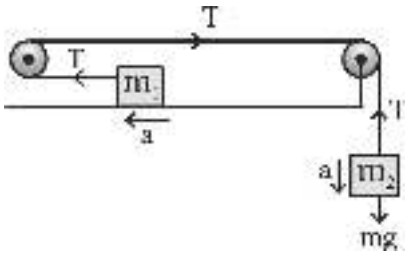
or $$\mu\left(\frac{2}{5}F\right) \geq mg$$

or $$F \geq \frac{5}{2\mu}mg$$

So, $$F_{min} = \left(\frac{5}{2\mu}\right)mg$$

11)

If a is acceleration of m_2, then from figure.



$m_2g - T = m_2a$ ...(i)

and $T = m_1a = 1a$ ...(ii)

From (i), $m_2g - T = m_2a$

$1g - m_1a = m_2a$

$1g = (m_1 + m_2)a = (1 + 1)\,a$

$a = \frac{g}{2} = \frac{10}{2} = 5 m/s^2$

$\Rightarrow$ $T = m_1a = 1 \times 5 = 5N$

12) **Question Explanation:** Evaluate statements about friction and potential energy.

Concept: Conservative and non-conservative forces.

Solution:

A. Statement-1: Frictional forces are **non-conservative** because they always dissipate energy as heat and are path-dependent. Therefore. **Statement-1 is false.**
B. Statement-2: Potential energy **cannot** be associated with friction because the force is path-dependent and leads to energy dissipation. **Statement-2 is false.**

Final Answer : Option (4) Statement-1 is false, Statement-2 is false.

13) Area between curve and displacement axis

$\frac{1}{2} \times (12 + 4) \times 10 = 80\ J$

In this time body acquire kinetic energy = $\frac{1}{2}mv^2$

by the law of conservation of energy

$\frac{1}{2}mv^2 = 80\,J$

$\Rightarrow \frac{1}{2} \times 0.1 \times v^2 = 80$

$\Rightarrow v^2 = 1600$

$\Rightarrow v = 40$ m/s

14) Work done by static friction on an object may be positive, negative and zero.

15) $w_g + w_a = \Delta k$

$-mgh - w_a = 0 - \frac{1}{2}mv^2$

$w_a = \frac{1}{2}mv^2 - mgh = \frac{1}{2} \times 1 \times 20 \times 20 - 1 \times 10 \times 18$

= 20 J

16)

Question Explanation:A particle slides down a frictionless track and moves horizontally at the end. We find the horizontal distance where it hits the ground.
Concept Based: Using energy conservation to find velocity at the end, and kinematics to determine horizontal displacement.
Formula used

Energy conservation : $mgh = \frac{1}{2}mv^2$

Time of flight t : = $\sqrt{\frac{2h}{g}}$

Horizontal distance x = vt
Calcualtion: From energy conservation :

$mg(1.0-0.5) = \frac{1}{2}mv^2$

$mg(0.5) = \frac{1}{2}mv^2$

Time of flight

$t = \sqrt{\frac{2(0.5)}{g}} = \sqrt{\frac{1}{g}}$

Horizontal distance

$x = vt = \sqrt{g} \times \sqrt{\frac{1}{g}} = 1m$

17)

By WET

$W_{friction} = \Delta KE$

$-\mu mgx = 0 - E$

$x = \frac{E}{\mu mg}$

18)

Given that $x = \frac{3t^2}{2}$

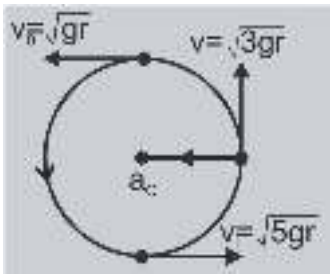
$v = \frac{dx}{dt} = 6t$

Velocity at t = 0 $v_i = 0$ m/s

at t = 1 $v_f = 6$ m/s

$W = \frac{1}{2}m\left[v_f^2 - v_i^2\right] = \frac{1}{2}(4)[(6)^2 - 0]$

= 72 J

19) 

$$a_C = \frac{v^2}{r} = \frac{3gr}{r}$$

$a_C = 3g$

20) $\alpha = t^2 - 2t + 2$·

$$\frac{d\omega}{dt} = t^2 - 2t + 2$$

$d\omega = (t^2 - 2t + 2)\, dt$

$$\int d\omega = \int \left(t^2 - 2t + 2\right) dt$$

$$\omega = \frac{t^3}{3} - \frac{2t^2}{2} + 2t$$

$$\omega = \frac{t^3}{3} - t^2 + 2t$$

$$\omega_{t=2\,sec} = \frac{8}{3} - 4 + 4 = \frac{8}{3}\ rad/s$$

21) $\theta = \theta_0 + \theta_1 t + \theta_2 t_0^2$

$$\frac{d\theta}{dt} = \omega = 0 + \theta_1 + \theta_2 (2t)$$

$$\frac{d\omega}{dt} = \alpha = 0 + \theta_2 (2)$$

$\alpha = 2\theta_2$ Independent of time, so a straight line parallel to y-axis

↓

$a_t = R\alpha$

$a_t = R(2\theta_2)$

a_t = constant (Independent of time)

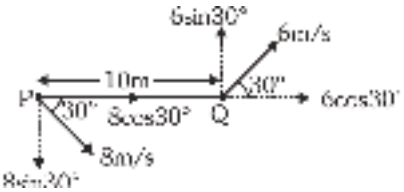
22) **Question Explanation :** Asking about direction of angular velocity given linear velocity direction.

Concept : $\vec{v} = \vec{\omega} \times \vec{r}$ (cross product). Right hand thumb rule

Solution :

Angular velocity is given by the right hand screw OR right hand thumb rule and your right hand fingers in the direction of circular motion, right hand's thumb will point in the direction of angular velocity Here in this case.

Final Answer : Option (3)



23) 8sin30°

$$\omega_{PQ} = \frac{(V_{PQ})_{\perp}}{r} = \frac{8\sin 30^\circ + 6\sin 30^\circ}{10}$$

ω_{PQ} = 0.7 rad/s

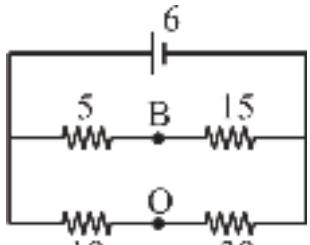
24) 4Ω and 6Ω resistances are short circuited. Therefore, no current will flow through these two resistances.
Current passing through the battery is

$$I = \left(\frac{20}{2}\right) = 10A$$

This is also the current passing in the wire AB from B to A.
Power supplied by the battery.
P = EI = (20)(10) = 200 watt
Potential difference across 4Ω resistance
= potential difference across 6Ω resistance = 0.

25) $$I_g = \left(\frac{S}{S+G}\right) I$$

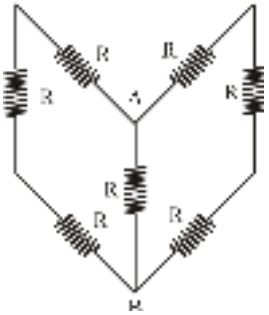
or $$\frac{I_g}{I} = \frac{S}{S+G} \text{ or } \frac{1}{34} = \frac{S}{S+G} \Rightarrow S = \frac{G}{33} = \frac{3663}{33} = 111\ \Omega$$



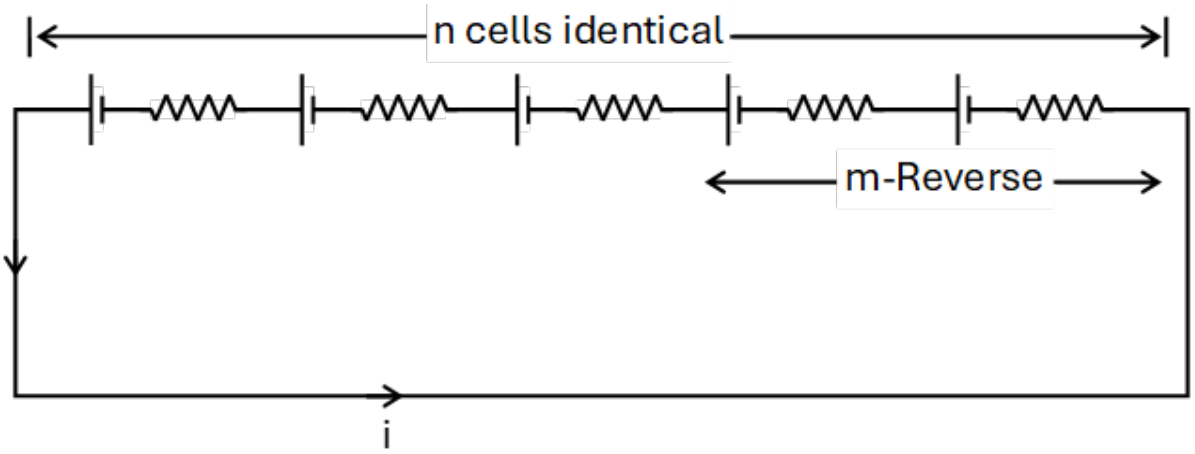
26) $V_B - V_D = 0$

27)

Effective circuit is



28) **Visual Diagram:**



net emf = (n – m)E – m(E) ⇒ (n – 2m)E

Calculation:

• Net emf: E_{net} = (6 – 2)E = 4E

• Total resistance: 6r

• Current: I = $\frac{4E}{6r} = \frac{2E}{3r}$

• Terminal voltage of reversed cell: V = E + Ir = E + $\frac{2E}{3} = \frac{5E}{3}$

Answer: $\frac{5}{3}$**E**

29)

According to the question :

$R_{series\ at\ 0\ °C} = R_{series\ at\ t\ °C}$

⇒ $R_{1\ at\ 0\ °C} + R_{2\ at\ 0\ °C} = R_{1\ at\ t\ °C} + R_{2\ at\ t\ °C}$

⇒ $R_{1\ at\ 0\ °C} + R_{2\ at\ 0\ °C} = R_{1\ at\ 0\ °C}[1 + \alpha_1.\Delta t] + R_{2\ at\ 0\ °C}[1 + \alpha_2.\Delta t]$

⇒ $R_{1\ at\ 0\ °C} + R_{2\ at\ 0\ °C} = R_{1\ at\ 0\ °C}[1 + (+\alpha).\Delta t] + R_{2\ at\ 0\ °C}[1 + (-3\alpha).\Delta t]$

⇒ $R_{1\ at\ 0\ °C} + R_{2\ at\ 0\ °C} = R_{1\ at\ 0\ °C}[1 + \alpha.\Delta t] + R_{2\ at\ 0\ °C}[1 - 3\alpha.\Delta t]$

⇒ $R_{1\ at\ 0\ °C} + R_{2\ at\ 0\ °C} = R_{1\ at\ 0\ °C} + R_{1\ at\ 0\ °C}.\alpha.\Delta t + R_{2\ at\ 0\ °C} - 3R_{2\ at\ 0\ °C}.\alpha.\Delta t$

⇒ $0 = R_{1\ at\ 0\ °C}.\alpha.\Delta t - 3R_{2\ at\ 0\ °C}.\alpha.\Delta t$

⇒ $-R_{1\ at\ 0\ °C}.\alpha.\Delta t = -3R_{2\ at\ 0\ °C}.\alpha.\Delta t$

⇒ $\frac{R_{1\ at\ 0\ °C}}{R_{2\ at\ 0\ °C}} = \frac{3}{1}$ = 3

Remark For Operator :- Answer :- Option (2.)
Question Level :- Medium

30) **Solution/Explanation/Calculation:**

When heater wires are connected in series then equivalent resistance.

$R_s = R + R + R = 3R$ ($\because$ wires are of same material and length :- $R_1 = R_2 = R_3 = R$)

⇒ Rate of heat produced in series connection

$$H_S = \frac{V^2}{R_3}$$

$$\Rightarrow H_S = \frac{V^2}{3R} \quad .....(1)$$

In second case (Parallel connection)

$$R_P = \frac{R}{3}$$ ($\because R_1 = R_2 = R_3 = R$)

 Heat produced in 2nd case

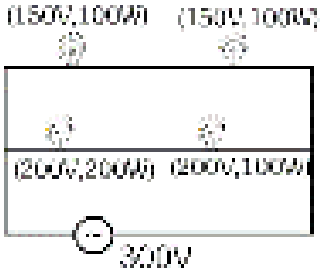
$$\Rightarrow H_P = \frac{V^2}{R_P} = \frac{V^2}{R/3} = \frac{3V^2}{R} \quad ..... (2)$$

$$\frac{H_S}{H_P} = \frac{V^2}{3R} \div \frac{3V^2}{R} = \frac{1}{9}$$

Hence, option (3) is correct.

31) **Problem Explanation :**

A. **Given ⇒** An electric circuit with an emf source and four electric bulbs is shown in the diagram :



A. **To find ⇒** The total power consumption by all four bulbs ⇒ P_{total} = *?*

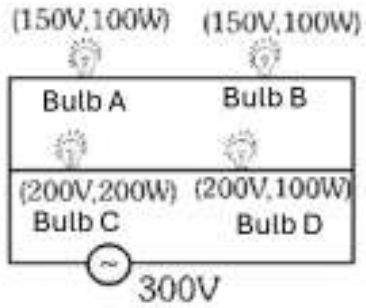
Solution :-

A. **Step-1 :**

B. **Concept ⇒** Calculation of the resistance of a bulb with the help of its rated voltage and rated power.

C. **Formula ⇒**
$$R = \frac{V_{rated}^2}{P_{rated}}$$

A. **Visual-Aid ⇒**



A. **Calculation ⇒**

The resistance of the bulb A

$$R_A = \frac{(150)^2}{100} = \frac{150 \times 150}{100} = 225\ \Omega$$

The resistance of the bulb B

$$R_B = \frac{(150)^2}{100} = \frac{150 \times 150}{100} = 225\ \Omega$$

The resistance of the bulb C

$$R_C = \frac{(200)^2}{200} = \frac{200 \times 200}{200} = 200\ \Omega$$

The resistance of the bulb D

$$R_D = \frac{(200)^2}{100} = \frac{200 \times 200}{100} = 400\ \Omega$$

A. **Step-2 :**

Concept ⇒ Distribution of the potential across all the bulbs (resistances).

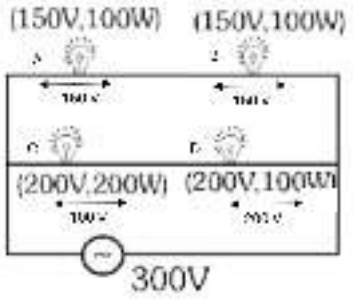
A. **Formula ⇒**

1. For parallel combination of the bulbs ⇒ The potential drop is the same across all.

2. For series combination of the bulbs ⇒ The potential drop is in the direct ratio of the resistors i.e.

$$V_1 : V_2 = R_1 : R_2$$

A. **Visual-Aid ⇒**



A. **Calculation ⇒**

The calculation of the potentials of the bulbs A & B

$$V_A : V_B = R_A : R_B$$

$$= 225 : 225$$

$$= 1 : 1$$

$$V_A = \frac{1}{1+1} \times 300 = \frac{1}{2} \times 300 = 150\,Volts$$

$$V_B = \frac{1}{1+1} \times 300 = \frac{1}{2} \times 300 = 150\,Volts$$

The calculation of the potentials of the bulbs C & D

$$V_C : V_D = R_C : R_D$$

$$= 200 : 400$$

$$= 1 : 2$$

$$V_C = \frac{1}{1+2} \times 300 = \frac{1}{3} \times 300 = 100\,Volts$$

$$V_D = \frac{2}{1+2} \times 300 = \frac{2}{3} \times 300 = 200\,Volts$$

A. **Step-3 :**

Concept ⇒ Calculation of the dissipated power of a bulb with the help of dropped voltage across it and its resistance.

A. **Formula ⇒** $P_{dissipated} = \frac{V^2_{dropped}}{R}$

A. **Calculation ⇒**

The dissipated power of the bulb A

$$P_A = \frac{(150)^2}{225} \quad \frac{150 \times 150}{225} \quad 100\,Watts$$

The dissipated power of the bulb B

$$P_B = \frac{(150)^2}{225} \qquad \frac{150 \times 150}{225} \qquad 100\ Watts$$

The dissipated power of the bulb C

$$P_C = \frac{(100)^2}{200} \qquad \frac{100 \times 100}{200\ \Omega} \qquad 50\ Watts$$

The dissipated power of the bulb D

$$P_D = \frac{(200)^2}{400} \qquad \frac{200 \times 200}{400} \qquad 100\ Watts$$

A. **Step-4 :**

Concept ⇒ Calculation of the total dissipated power of the circuit.

A. **Formula ⇒** $P_{total} = P_A + P_B + P_C + P_D$

Calculation ⇒ The total dissipated power of the given circuit :-

$P_{total} = 100 + 100 + 50 + 100 \Rightarrow P_{total} = 350\ Watts$

Answer :- Option (2.)

Question Level :- Medium

32)

A. $H=V^2/R$ (for parallel)
B. $H=I^2 \times R$ (for series)
C. V^2= Power × R

Calculation:
Given: $H_{1\,s}$=120cal/s,R_1 s = 150
V^2 =120 × 15 = 1800
14Ω and 16Ω are in series - R_total =30Ω
Same V across both branches
Current through series: I = V / 30
Heat in 140: $H_{14}=V^2/900\times14=(1800/900)\times14=2\times14=28$cal/s

33)

Conceptual

34) $R_{eq} = \frac{2Rr}{R+r}$

$i_{Total} = \frac{E}{R_{eq}} = \frac{E(R+r)}{2Rr}$

$i_{AB} = \frac{r}{R+r} i_{Total}$

$i_{AC} = \frac{R}{R+r} i_{Total}$

$i_{BC} = i_{AC} - i_{AB}$ (from C to B)

$= \frac{R-r}{R+r} i_{Total}$

$= \frac{R-r}{R+r}\left(\frac{E(R+r)}{2Rr}\right) = \frac{E(R-r)}{2Rr}$

35) $T' = 2\pi\sqrt{\frac{I'}{M'B}}$

Calculation :

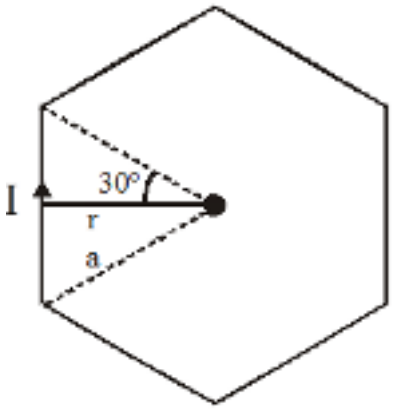
Substituting I' = 3I and M' = M/3:

$T' = 2\pi\sqrt{\frac{3I}{(M/3)B}}$

$T' = 2\pi\sqrt{\frac{3}{1/3}\cdot\frac{1}{MB}}$

$T' = 2\pi\sqrt{9}\sqrt{\frac{1}{MB}}$

$T' = 3T_0$



36) r = acos30°

$B = \frac{6\mu_0 I}{4\pi a\cos 30^\circ} \times 2\sin 30^\circ \times 50$

$= \frac{\mu_0 I}{\pi}\frac{150}{\sqrt{3}a} = \frac{50\sqrt{3}}{0.1}\frac{\mu_0 I}{\pi}$

$= 500\sqrt{3}\frac{\mu_0 I}{\pi}$

37) $m = NIA = 1 \times I \times a^2$

here a = side of square

Now,

$4a = 2\pi r$

$r = \frac{2a}{\pi}$

For circular loop

$m' = 1 \times I \times \pi r^2$

$= 1 \times I \times \pi \times \left(\frac{2a}{\pi}\right)^2$

$m' = \frac{4m}{\pi}$

38) For permanent magnet we prefer a material with high retentivity (so as to make a stronger magnet) and high coercivity (so that magnetization any not be wiped out easily). for electromagnet we prefer high saturated magnetism low coercivity and least possible area of hysteresis loop so that electromagnet develops high magnetization, is easily demagnetized and energy loss in a magnetization cycle is least. Therefore, P is suitable for making permanent magnet and Q for making electromagnet.

39) Magnetic field in first quadrant at point (x, y)

$\vec{B} = \frac{\mu_0 I}{2\pi y} + \frac{\mu_0 (3I)}{2\pi x} = \vec{0} \Rightarrow x = 3y$

40)

Since $\vec{a} \perp \vec{B} \Rightarrow \vec{a} \cdot \vec{B} = 0$

$\Rightarrow -10 + 3a + a^2 = 0$

$\Rightarrow (a - 2)(a + 5) = 0$

$\Rightarrow a = 2, -5$

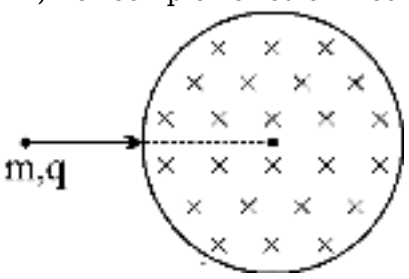
41) Length PS = $\sqrt{2}r$ {r = radius of circular part of loop}

$F_{PS} = i\vec{l}B = i\sqrt{2}rB = F$

length PQR = 2 r

$F_{PQR} = \sqrt{2}F$

42) For complex circular motion



$T = \frac{2\pi m}{qB}$

Since it is deflection by 60°

$t = T\left(\frac{2}{2\pi}\right) = \frac{2\pi m}{qB}\frac{\pi}{3 \times 2\pi} = \frac{\pi m}{3qB}$

43)

For +ve ions.

Force due to electric field is along + x

Force due to mag. field

$\vec{F} = q[\vec{V} \times \vec{B}]$

$\vec{F} = (+q)\left[V\hat{i} \times B\hat{k}\right]$

$\vec{F} = qvB(-\hat{j})$

So, positive ions get deflected in -y dir.

For -ve ions.

Force due to electric field is along -x.

Force due to mag. field

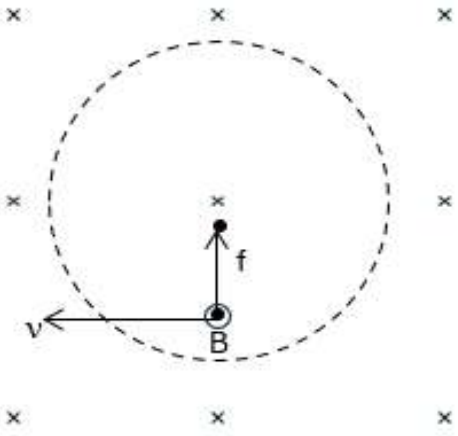
$\vec{F} = q\left[\vec{V} \times \vec{B}\right]$

$= -q\left[-v\hat{i} \times B\hat{k}\right]$

$= qvB(-\hat{j})$

So, negative ions get deflected in - y din.

Answer : (2)



44)

Solution :

• Magnetic force : $F = e\nu B$

• Centripetal force : $F = \frac{m\nu^2}{r}$

• Equate both forces : $e\nu B = \frac{m\nu^2}{r}$

• Simplify : $\frac{e}{m} = \frac{\nu}{rB}$

Hence,

Option (1) is correct : $\frac{v}{Br}$

45) Magnetic field due to wire (1) & (2) is zero, because point P lies on the axis of these wires

Magnetic field due to wire (3)

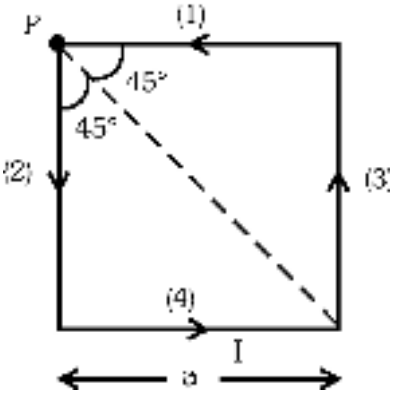
$B_3 = \frac{\mu_0 I}{4\pi a}(\sin 0^\circ + \sin 45^\circ) = \frac{\mu_0 I}{4\sqrt{2}\pi a}$

Magnetic field due to wire (4)

$B_4 = \frac{\mu_0 I}{4\pi a}(\sin 0^\circ + \sin 45^\circ) = \frac{\mu_0 I}{4\sqrt{2}\pi a}$

Net magnetic field due to wire (3) and (4) :-

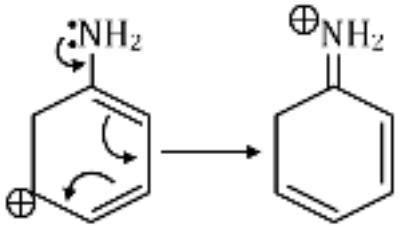
$B_p = B_3 + B_4 = 2 \cdot \frac{\mu_0 I}{4\sqrt{2}\pi a} = \frac{\mu_0 I}{2\sqrt{2}\pi a}$



CHEMISTRY

46)

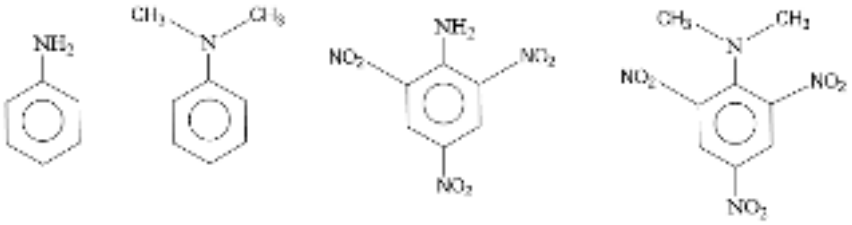
Question Explanation: Stability of carbocations depends on resonance, hyperconjugation, and structure.
Concept: Stability order of Carbocations
Solution: Carbocation stability is primarily governed by **Resonance >Hyperconjugation >Inductive Effect**.
5 (most stable) — The positive center is **allylic/benzylic** to a ring double bond so it is strongly resonance-stabilized, and it also has better alkyl substitution/hyperconjugation than (4).
4 — Also **allylic/benzylic** (good resonance stabilization) but lacks the extra hyperconjugative support that (5) has, so slightly less stable.
2 — Located on the 3° carbocation where it gets hyperconjugation stabilization with **no resonance** and more alkyl substitution than (1), so intermediate stability.
1 — Secondary center where lesser hyperconjugation than third one with **no resonance** delocalisation (only hyperconjugation), so less stable than 2.
3 (least stable) — Essentially a **primary carbocation** with **no resonance**, so the weakest stabilization.
So, correct stability order for given carbocations will be **5 > 4 > 2 > 1 > 3**
Final Answer: (Option 1).

47) 

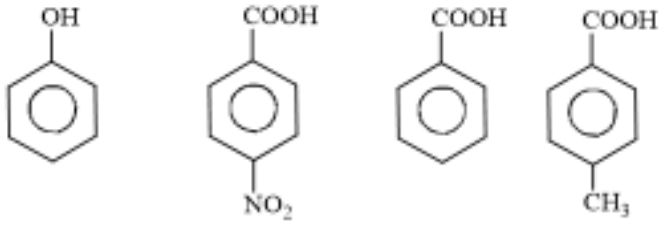
The +M effect of NH_2 is stabilizing the carbocation.

48)

Conceptual



49) Steric factor



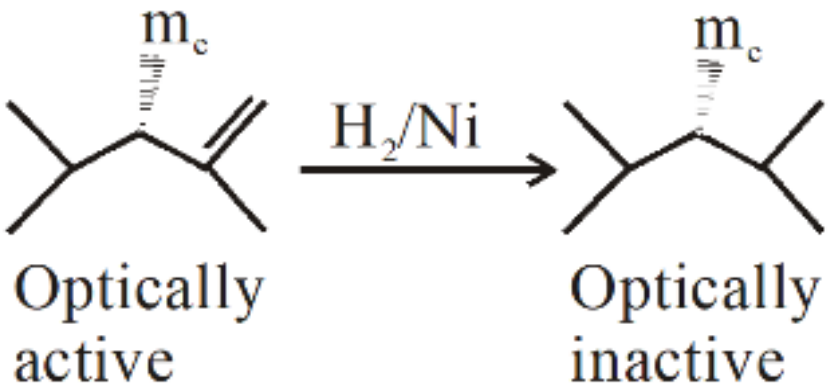
50) II > III > IV > I

51)

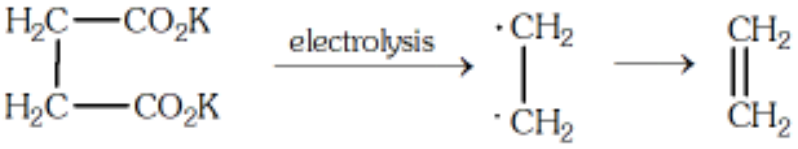
Conceptual

52)

NBS is used for the bromo substitution of allylic hydrogen.



53)

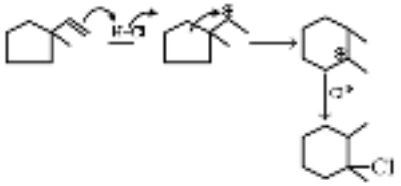


54)

55) Rate of electrophillic addition µ nucleophilicity of alkene

56)

HCl does not show peroxide effect



57)

Conceptual

58)

Conceptual

59)

Conceptual

60)

Conceptual

61)

Explanation

A. Linear shape occurs when the central atom has two bond pairs and zero lone pairs, resulting in a bond angle of 180°.

Concept

A. XeF_2: Xe has 8 valence electrons. It forms 2 bonds with F, using 2 electrons. This leaves 6 non-bonding electrons or 3 lone pairs. Arrangement is trigonal bipyramidal, but shape is linear (F-Xe-F) due to lone pairs occupying equatorial positions.

B. I_3^-: The central I has 7 valence electrons. It forms 2 bonds with other I atoms and has a negative charge (1 extra electron), totaling 10 electrons. This means 2 bond pairs and 3 lone pairs. Arrangement is trigonal bipyramidal, but shape is linear (I-I-I) due to lone pairs occupying equatorial positions.

C. CO_2: C has 4 valence electrons. It forms 2 double bonds with O atoms, using all 4 electrons. This means 2 bond pairs and zero lone pairs. Shape is linear (O=C=O).

Answer option 4,(All).

62)

fact

63) $\underset{}{H-\overset{\oplus}{\underset{sp}{N}}=\overset{\ominus}{\underset{sp}{N}}-H} \rightarrow 50\%$ s-character

64) Lone pair on equitorial position.

65)

Asking for:

Identify the linear species among N_3^-, I_3^+, I_3^-, and BrF_3.

Concept:

A. **VSEPR Theory (Valence Shell Electron Pair Repulsion Theory):** Predicts the geometry of molecules based on electron pair repulsion.
B. **Steric Number:** The number of atoms bonded to a central atom plus the number of lone pairs on the central atom.
C. **Linear Geometry:** Occurs when the central atom has a steric number of 2 with no lone pairs, or a steric number of 3 with 2 lone pairs, or a steric number of 5 with 3 lone pairs.

Formula:

A. Steric Number = Number of bonded atoms + Number of lone pairs

Solution/Explanation:

A. **N_3^- (Azide Ion):**
    A. Total valence electrons: 3 * 5 (from N) + 1 (from negative charge) = 16
    B. Lewis structure: $[:N=N=N:]^-$
    C. Central N atom has 2 bonding pairs and no lone pairs.
    D. Steric number = 2
    E. Shape: Linear
B. **I_3^+ (Triiodide Cation):**
    A. Total valence electrons: 3 * 7 (from I) - 1 (from positive charge) = 20
    B. Lewis structure: $[I-I-I]^+$ (with two lone pairs on the central iodine)
    C. Central I atom has 2 bonding pairs and 2 lone pairs.
    D. Steric number = 4
    E. Shape: Bent (not linear)
C. **I_3^- (Triiodide Anion):**
    A. Total valence electrons: 3 * 7 (from I) + 1 (from negative charge) = 22
    B. Lewis structure: $[I-I-I]^-$ (with three lone pairs on the central iodine)
    C. Central I atom has 2 bonding pairs and 3 lone pairs.
    D. Steric number = 5
    E. Shape: Linear
D. **BrF_3 (Bromine Trifluoride):**
    A. Total valence electrons: 7 (from Br) + 3 * 7 (from F) = 28
    B. Lewis structure: Br with 3 F atoms and 2 lone pairs.
    C. Central Br atom has 3 bonding pairs and 2 lone pairs.
    D. Steric number = 5
    E. Shape: T-shaped (not linear)

Therefore, the linear species are N_3^- and I_3^-.

Answer : Option : 3

66)

Explanation

A. Bond energy is the energy required to break a chemical bond. Generally, stronger bonds have higher bond energies. We need to match the given bonds with their approximate bond energies.

Concept

A. O–O: This is a relatively weak single bond due to lone pair repulsion. Its bond energy is around 146 kJ mol^{-1} (q).

B. O–H: This is a strong polar bond. Its bond energy is around 467 kJ mol^{-1} (p).

C. S–S: This single bond is stronger than O–O due to less lone pair repulsion. Its bond energy is around 266 kJ mol^{-1} (s).

D. S–H: This bond is weaker than O–H but stronger than S–S. Its bond energy is around 374 kJ mol^{-1} (r).

Correct Match:

A. O–O - (q) 146 kJ mol^{-1}

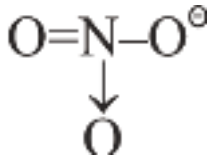
B. O–H - (p) 467 kJ mol^{-1}

C. S–S - (s) 266 kJ mol^{-1} (

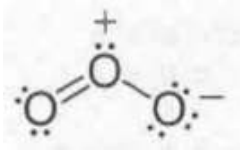
D. S–H - (r) 374 kJ mol^{-1}

Answer option 1, (i-q, ii-p, iii-s, iv-r).

67) Size ↑ repulsion ↑ B.A. ↑

68) 

69) **Answer - Option(4)**
Explanation -
Concept - Here's how to calculate formal charges:
1. Lewis Structure: Ozone (O_3) has a bent structure. One oxygen atom is central, single-bonded to another oxygen, and double-bonded to the third. There is also a coordinate bond (both electrons from one atom) from one of the terminal oxygens to the central oxygen.



2. Formal Charge Formula: Formal Charge = (Valence Electrons) - (Non-bonding Electrons) - ($\frac{1}{2}$ Bonding Electrons)

3.Calculations:

A. Central Oxygen: 6 (valence electrons) - 2 (non-bonding electrons) - ($\frac{1}{2}$ × 6 bonding electrons) = +1

B. Oxygen with Single Bond: 6 (valence electrons) - 6 (non-bonding electrons) - ($\frac{1}{2}$ × 2 bonding electrons) = -1

C. Oxygen with Double Bond: 6 (valence electrons) - 4 (non-bonding electrons) - ($\frac{1}{2}$ × 4 bonding electrons) = 0

Therefore, the formal charges are 0, +1, and -1.

70)

$\mu = CH_3Cl > CH_3F > CH_3Br > CH_3I$

71) **Explanation** -
The question asks you to determine the number of antibonding electrons present in the molecular orbitals of three oxygen species
Concept - (Number of antibonding electron in O_2, $O^{\ominus}_2$ & O_2^{2-})
O_2^- : $(\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2 (\sigma^* 2s)^2 (\sigma 2pz)^2 (\pi 2px)^2 (\pi 2py)^2 (\pi^* 2px)^2 (\pi^* 2py)^1$

A. Antibonding electrons: 7

O_2 : $(\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2 (\sigma^* 2s)^2 (\sigma 2pz)^2 (\pi 2px)^2 (\pi 2py)^2 (\pi^* 2px)^1 (\pi^* 2py)^1$

A. Antibonding electrons: 6

O_2^{2-} : $(\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2 (\sigma^* 2s)^2 (\sigma 2pz)^2 (\pi 2px)^2 (\pi 2py)^2 (\pi^* 2px)^2 (\pi^* 2py)^2$

A. Antibonding electrons: 8

Final Answer - Option (3)

72) **EXPLANATION:-** correct order of increasing bond order

Concept Used:

Bond order (B.O.) =BONDING ELECTRON-ANTI BONDING ELECTRON/2

SOLUTION:-

1. He_2^+

B.O. = (2 – 1)/2 = 0.5

2. O_2^-
B.O. = (10 – 7)/2 = 1.5

3. NO

B.O. = (10 – 5)/2 = **2.5**

4. C_2^{2-}
B.O. = (10 – 4)/2 = 3

$He_2^+ < O_2^- < NO < C_2^{2-}$

Final Answer : Option (2)

73) Due to different No. of crystalline water.

74) Metallic character ∝ Ionic nature

75)

Explain Question: Check validity of statement about octet expansion in 2nd-period elements.

Concept: octet rule, covalency.

Solution: 2nd-period elements (Li–Ne) lack 2d orbitals, so they cannot expand octet. Thus, both assertion and reason are true, and reason explains assertion correctly.

Final Answer: Option (3)

76)

Ans is (A)

$$[H_2CO_3] = \frac{0.1}{0.1} = 1M \qquad Ka_1 = 4 \times 10^{-6}, \quad Ka_2 = 5 \times 10^{-11}$$

$$\underset{1M}{\underset{1-\alpha_1 M}{H_2CO_3}} \rightleftharpoons \underset{\alpha_1 M+\alpha_1+\alpha_2 M}{H^+} + \underset{\alpha_1 M=\alpha_1\alpha_2=\alpha_1 M}{HCO_3^-}$$

$$\underset{\alpha_1(1-\alpha_2)M=\alpha_1 M}{HCO_3^-} \rightleftharpoons \underset{\alpha_1+\alpha_1\alpha_2 M=\alpha_1 M}{H^+} + \underset{\alpha_1\alpha_2 M}{CO_3^{2-}}$$

Now, $$Ka_1 = \frac{[H^+][HCO_3^-]}{[H_2CO_3]} = \frac{\alpha_1 \times \alpha_1}{1-\alpha_1} \Rightarrow 4 \times 10^{-6} = \frac{\alpha_1^2}{1-\alpha_1}$$

$$\therefore \ \alpha_1 = 2 \times 10^{-3} M$$

$$Ka_2 = \frac{[H^+]\left[CO_3^{2-}\right]}{[HCO_3^-]} = \frac{\alpha_1 \times \left[CO_3^{2-}\right]}{\alpha_1} \Rightarrow \left[CO_3^{2-}\right] = 5 \times 10^{-11} M$$

$$\therefore \ [H^+] = 2 \times 10^{-3}, [HCO_3^-] = 2 \times 10^{-3} M \quad \left[CO_3^{2-}\right] = 5 \times 10^{-11} M$$

$$\therefore \ [OH^-] = \frac{10^{-14}}{2 \times 10^{-3}} = 0.5 \times 10^{-11} = 5 \times 10^{-12} M$$

77) Combination of weak base (NH_4OH) and its salt with strong acid (NH_4Cl) act as basic buffer solution

$$\underset{\text{(weak base)}}{NH_4OH} + \underset{\text{(Strong acid)}}{HCl} \rightleftharpoons NH_4Cl + H_2O$$

initial	2 mole	1 mole	0	0
After reaction	$(2-1)$ =1mole	$(1-1)$ = 0	1 mole	1 mole

1 mole

NH_4OH & 1 mole NH_4Cl are present in mixture after reaction. Hence it will also act as basic buffer

78) According to bronsted-lowry concept, proton donor act as an acid

79) $pH = pKa + \log \frac{[CH_3COO^-]}{[CH_3COOH]}$

$5.18 = (5 - \log 2) + \log \frac{[CH_3COO^-]}{[CH_3COOH]}$

$5.18 - 4.7 = 0.48 = \log 3 = \log \frac{[CH_3COO^-]}{[CH_3COOH]}$

$\therefore \frac{[CH_3COO^-]}{[CH_3COOH]} = 3 : 1$

80) $s = 6.9 \times 10^{-2}$ gm/100 ml

$= \frac{6.9 \times 10^{-2}}{690} = 10^{-4}$ mol/100 ml

$= 10^{-3}$ mol/lit

$K_{sp} = 108\ s^5 = 108 \times 10^{-15} = 1.08 \times 10^{-13}$

81)

C < B < A < D

in NH_3 solubility increases due to formation of complex.

$Ag^+ + 2NH_3 \rightarrow [Ag(NH_3)_2]^+$ in NaCl & $BaCl_2$

solubility decreases due to increase in conc. of $[Cl^-]$

82) $Fe^{2+} \xrightarrow{1e^-} Fe^{3+}$

$C^{+2} \xrightarrow{12e^-} C^{+4}$ (Total e^- lost = 61)

$N^{-3} \xrightarrow{5e^-} N^{+5}$

$Mn^{+7} \xrightarrow{5e^-} Mn^{2+}$ (Total e^- gained = 5)

$\therefore 5[Fe(CN)_6]^{4+} + 61MnO_4^-$

5 mole 61 moles

? 1 mole

$\therefore \frac{5}{61}$ moles are required $Mn^{+7} \xrightarrow{5e^-} Mn^{2+}$

83) H_2CO_3/HCO_3^-

84) $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

initially 1 3 0

at eq^n (1–0.5) (3–1.5) 2×0.5 = 1 mole

$\Rightarrow$ Total moles at equilibrium = 0.5 + 1.5 + 1 = 3

Partial pressure of NH_3 = $P \times \left(\frac{1}{3}\right) = \frac{P}{3}$

85) K_c depends only temperature.

86)

Conceptual

87) $H_2(g) + S(s) \rightleftharpoons H_2S(g)$

$$\log \frac{K_2}{K_1} = \frac{\Delta H}{2.303R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

$$\log \left(\frac{9.25}{18.5} = \frac{1}{2}\right) = \frac{\Delta H}{2.303 \times 8.314}\left(\frac{1}{925} - \frac{1}{1000}\right)$$

$$-0.3010 = \frac{\Delta H}{2.30 \times 8.314}\left(\frac{75}{925 \times 1000}\right)$$

ΔH = – 71080.57 J/mole

88) Moles at eqm $\underset{2-x}{A_2(g)} + \underset{4-x}{B_2(g)} \rightleftharpoons \underset{2x}{2AB(g)}$

$$K_c = \frac{4x^2}{(2-x)(4-x)}$$

$$\Rightarrow x = \frac{32}{24} = 1.33 \text{ mole}$$

$$[AB(g)] = \frac{2 \times 1.33}{4} = 0.66\ M$$

89)

Element	%	Mole	Mole Ratio
X_e	53.5	$\frac{53.5}{131} = 0.4$	$\frac{0.4}{0.4} = 1$
F	46.5	$\frac{46.5}{19} = 2.4$	$\frac{2.4}{0.4} = 6$

Hence compound may be XeF_6.

90) $2\overset{+2}{N}O + 3\overset{0}{I_2} + 4H_2O \rightarrow 2\overset{+5}{N}O_3^- + 6\overset{-1}{I}^- + 8H^+$

BIOLOGY

91)

Question Explanation: Ground tissue layer differentiation.
Concept: Plant Anatomy.
Solution: The ground tissue in dicot roots and stems and monocot roots is organized into distinct layers like the cortex and pith. However, in monocot stems, the ground tissue is not differentiated into these layers. Instead, it's a uniform, undifferentiated mass of parenchyma cells that fills the entire cross-section of the stem.
Final Answer: Monocot stem.

92) **Question Explanation**: Features of dicot stem vascular tissue.
Concept: Anatomy of Dicot Stem
Solution: The vascular bundles of a **dicotyledonous stem** are **conjoint** (xylem and phloem on the same radius), **collateral** (phloem is on the outer side of xylem), and **open** (they have a cambium). The xylem is also **endarch**, with protoxylem towards the pith.
Final Answer: Conjoint, collateral, open, endarch

93)

NCERT (XI) Pg. # 95 (Fig.)

94)

NCERT (XI) Pg. # 96

95) NCERT XI, Pg. No.93

96) NCERT Page No. # 91, 93, 94

97) **Solution:** Those cells which are formed towards the outside of cork cambium become suberized. Due to this, these cells become dead. These dead cells are known as cork phellem.

98)

Dicot stem and Monocot stem

99)

NCERT-XI, Pg # 94

100)

NCERT XI Page No. 93

101)

NCERT XI, Pg. No. 85

102)

NCERT XI, Pg. No. 89

103)

Correct answer: Option 2 – a → iv, c → v, e → i

EXPLANATION-

match each correctly:

A. (a) Jan Ingenhousz → (iv) Essentiality of sunlight to plants
   He showed that sunlight is necessary for plants to release oxygen.
B. (b) F.F. Blackman → (iii) Law of limiting factor
   He proposed that photosynthesis is limited by the factor in least supply.
C. (c) Joseph Priestley → (v) Essential role of air in the growth of green plants
   He discovered that plants modify air (oxygen production).
D. (d) Julius von Sachs → (ii) *(should be: Demonstrated that glucose is formed and stored as starch in green parts of plant)* – not listed in options.
E. (e) T.W. Engelmann → (i) First action spectrum of photosynthesis
   He used filamentous algae and aerobic bacteria to map the most effective light wavelengths.

104) Module 3 Pg # 192

105) **Explain Question :** The question asks to identify the scientist who used purple and green bacteria to prove that photosynthesis is a light-dependent process where a hydrogen donor reduces carbon dioxide to carbohydrates.
Concept : This question is based on Early Experiments
Explanation : "Cornelius van Niel (1931) was the first scientist to demonstrate that photosynthesis is essentially a light-dependent reaction in which hydrogen from a suitable oxidisable compound reduces carbon dioxide to carbohydrates."
Final answer : 3

106) NCERT-XI Eng. Pg. # 210 (Figure 13.3(C) and last para)
NCERT-XI हिन्दी पेज # 210 (चित्र13.3(l) एवं पेज # 211 (दूसरा पेराग्राफ)

107) **Explain Question :** The question asks to identify the color of light at which the photosynthetic pigment chlorophyll-a absorbs the most energy.

Concept : absorption spectrum
Explanation : "Chlorophyll 'a' shows maximum absorption in the blue and the red regions (High, but less than blue)."
Final answer : 1

108)

A. Statement I: "CO_2 is the major limiting factor for photosynthesis" - This is correct. During photosynthesis, CO_2 is essential for the production of glucose. In many conditions, the concentration of CO_2 can limit the rate of photosynthesis.
B. Statement II: "C_4 plants adapted to dry tropical regions" This is also correct. C_4 plants, like maize and sugarcane, are adapted to hot and dry climates. They have a specialized pathway for photosynthesis that helps them efficiently use CO_2 and minimize water loss, making them well-suited for such environments.

Thus, the correct answer is:
4. Both statements I and II are correct.

109) From module

110)

NCERT-XI, Pg # 149,150

111) NCERT XI (E) Pg.# 144 (fig. 11.8) and pg # 143 (last para)
हिन्दी पेज # 144 (fp= 11.8) एवं पेज # 143 (अंतिम पेराग्राफ)

112)

NCERT XIth Pg. No. # 233

113)

NCERT (XIth) (E) Pg. # 220 (Photorespiration ⇒ 3rd Para)

NCERT (XIth) हिन्दी पेज # 220 (आखिरी पेराग्राफ)

114) NCERT XI Pg. # 219

115)

NCERT-XI Pg#146

116) NCERT (XIth) Pg. # 220

117)

The correct answer is: A proton pump, a membrane, a proton gradient, and ATPase.
Here's why:

A. **Proton pump:** This is responsible for actively transporting protons (H^+) across the membrane, creating a proton gradient.
B. **Membrane:** The membrane is essential for separating the compartments where the proton gradient is established and where ATP synthesis occurs.
C. **Proton gradient:** The gradient of protons across the membrane (more protons on one side) stores potential energy.
D. **ATPase:** The ATP synthase enzyme (often referred to as the Fo-F_1 complex) uses the energy from the movement of protons down their concentration gradient to synthesize ATP.

118) NCERT-XI, English Page No. # 213 & 214 (Topic - Chemiosmosis hypothesis)
हिन्दी पेज # 214 [बिन्दु (अ), (ब) एवं(स)]

119) NCERT XI Pg # 215

120) NCERT XI Pg. # 220

121) NCERT XI Eng. Page # 220
हिन्दी पेज # 220,222

122) NCERT, Pg. # 220

123) NCERT Pg # 142

124) NCERT Pg # 146

125)

The correct answer is:
$NADP^+$.
In non-cyclic photophosphorylation, the final electron acceptor is $NADP^+$ (Nicotinamide adenine dinucleotide phosphate). The electrons are transferred from the photosystem II to the electron transport chain, eventually reaching $NADP^+$, which combines with protons (H^+) to form NADPH. This NADPH is then used in the Calvin cycle to help synthesize glucose.

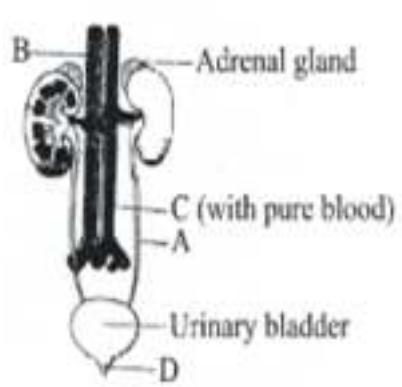
126) NCERT Pg. No. # 205

127)

Explain Question : Identify A to D structures in the given diagram.

Concept : This question is based on Human excretory system

Explanation :



Final answer : Option 3. Ureter – Inferior vena cava – Dorsal aorta – Urethra

128) **Explain the question :** This question is asking to match excretory functions with the respective parts of the human excretory system.
Concept : This question is based on urine formation and urinary tract.
Solution :
(i) Ultrafiltration - malpighian corpuscles (d)
(ii) Concentration of urine - henle's loop (a)
(iii) Transport of urine - ureter (b)
(iv) Storage of urine - urinary bladder (c)
Final answer : 1

129) NCERT Pg. # 297, Para-13

130) NCERT-XI, Pg. # 294

131) **Explain the question :** This question asking about absorption of different materials in specific parts of nephron.
Concept : This question is based on tubular reabsorption.
Solution :
(1) Ascending limb of LOH - NaCl reabsorbed - correct
(2) Descending limb of LOH - water reabsorbed - correct
(3) PCT - Glucose reabsorbed - correct
(4) DCT - amino acids are not reabsorbed here they are absorbed in PCT - incorrect.
Final answer : option (4)

132) NCERT, Pg. # 293

133) NCERT-XII, Pg. # 213

134) NCERT Pg. # 210,211

135) NCERT Pg. # 210,211

136) NCERT Pg. # 210,211

137) NCERT Pg. # 208

138) NCERT Pg.#211,212,213

139) NCERT Pg. # 211,212,213

140) NCERT Pg. # 213

141)

Question Explanation: Identify the correct structural or functional feature of muscle fibers.

Concept: The question is based on structure of contractile protein.

SOLUTION/EXPLAINATION:

Z-lines anchor myosin (thick) filament to the ends of the sarcomere:
This is incorrect. Z-lines anchor **actin (thin) filaments**, not myosin, to the ends of the sarcomere. Myosin is located in the A-band of the sarcomere.
1. Sarcoplasmic reticulum stores acetylcholine:
This is incorrect. The sarcoplasmic reticulum stores **calcium ions (Ca^{2+})**, not acetylcholine. Acetylcholine is a neurotransmitter stored in synaptic vesicles in the nerve terminals.
2. Myosin cross bridges contain calcium binding sites:
This is incorrect. Myosin cross bridges have **ATPase activity** for energy release and binding sites for actin, but calcium binding sites are located on **troponin**, a regulatory protein in thin filaments.
3. Actin and regulatory proteins are located in thin filament:
This is correct. The thin filament consists of **actin**, **tropomyosin**, and **troponin**. These regulatory proteins play a crucial role in muscle contraction.

Final answer: Actin and regulatory proteins are located in thin filament.

142)

4
NCERT Pg No - 224, 226, 227

143)

Question explanation: Identify which events occur during muscular contraction.

Concept: Sliding filament theory of muscle contraction

Solution:

(a) Correct - 'H' zone disappears – actin filaments slide inward.

(b) ‘A’ band widens – incorrect; A band remains constant.

(c) Correct - ‘I’ band reduces – actin slides inward.

(d) Correct - Myosin hydrolyzes ATP – provides energy for contraction.

(e) Correct - Z-lines pulled inward – sarcomere shortens.

Final Answer: – (a), (c), (d), (e) only

144)

Column-I (Types of Joints)		**Column-II (Locations)**	
a	Ball and socket joint	**ii**.	Between humerus and pectoral girdle
b	Hinge joint	**iv.**	Knee joint
c	Pivot joint	**iii.**	Between atlas and axis
d	Saddle joint	**i.**	Between carpal and metacarpal of thumb

Therefore, the correct answer is Option 2: a-ii, b-iv, c-iii, d-i.

145) NCERT Page#306.

146) **Question explanation:** This question is asking about matching different skeletal and muscular disorders with their correct causes or characteristics.

Concept : Disorders of the skeletal and muscular system

Explanation : Gout → Accumulation of uric acid crystals (iii)Osteoporosis → Decreased levels of estrogen (i)Tetany → Low Ca^{2+} ions in blood (ii)Muscular dystrophy → Genetic disorder (v)

Final answer : 2 — (a)-(iii), (b)-(i), (c)-(ii), (d)-(v)

147)

Explain Question: Identify the correct statement related to types of ribs in humans.

Concept: Rib cage – Classification of ribs.

Solution:

A. **Vertebrosternal ribs (True ribs):** These seven pairs of ribs are directly connected to the sternum by their costal cartilages.
B. **Vertebrochondral ribs (False ribs):** These three pairs of ribs (8th, 9th, and 10th) are indirectly connected to the sternum. They are connected to the cartilage of the ribs above them.
C. **Vertebral ribs (Floating ribs):** The last two pairs of ribs (11th and 12th) are not connected to the sternum at all; they are free at their anterior ends.

Therefore, the correct statement is that there are seven pairs of vertebrosternal, three pairs of

vertebrochondral, and two pairs of vertebral ribs.

Final Answer:
There are seven pairs of vertebrosternal, three pairs of vertebrochondral and two pairs of vertebral ribs.

148)

Explain Question : Which structure forms a bridge with actin during muscle contraction.

Concept : This question is based on Mechanism of muscle contraction.

Solution : The **head of myosin** binds to the exposed active site on actin to form the **cross bridge**, initiating contraction.

Final Answer : option (3)

149)

NCERT Pg.# 302

150)

Question Explanation : Choose the correct logical sequence distinguishing movement and locomotion.

Concept : The question is based on movement and locomotion.

Solution : The correct answer is option 2.

A. All **locomotion** are movements, but all **movement** are not **locomotion**.

Final answer : Option (2)

151)

NCERT (XI) Pg. # 307, para-20.2.2

152)

Question Explanation : The question is asking to identify which statements about the human skeleton are true or false.
Concept : Human skeleton system.
Solution :
(A) True, humans have 206 bones.
(B) True, humans have 12 pairs of ribs.
(C) False, sternum is on the ventral (anterior) side.
(D) True, most mammals have 7 cervical vertebrae, so statement is False.

(E) True, ribs have two articulation surfaces → bicephalic.
Final Answer : Option (3)

153) NCERT XI Pg # 306

154) Module-5, Pg. No. # 109

155)

Question Explanation: The question asks which canal passes through the midbrain.

Concept: midbrain

Solution: The cerebral aqueduct connects the third and fourth ventricles and passes through the midbrain.

Final Answer: Cerebral aqueduct

156)

157)

Question Explanation: The question is asking which structures together form the brain stem.

Concept: Components of the brain stem

Solution: The brain stem is composed of mid brain, pons and medulla oblongata, which connect the brain with the spinal cord and control vital functions.

Final Answer: Mid brain, pons and medulla oblongata.

158)

NCERT XI, Pg. # 321

159)

NCERT XI, Pg. # 321

160)

NCERT XI, Pg. No. 321, Para - 21.4.2

161) NCERT Pg.#322

162) NCERT Pg.#322

163) NCERT Pg.#322

164)

NCERT Based

165) **Question Explanation :** Activity of Na+-K+ pump during action potential

Concept: sub topic : Action Potential Phases

Solution:

• B (-55mV), C (0mV), and D (+10mV) represent states where the membrane potential is depolarized. During depolarization, other channels (like voltage-gated sodium channels) are opening and driving the membrane potential away from the resting potential. The Na+/K+ pump is still functioning at these potentials, but it's having to work against these other forces.

Final Answer : Option (3)

166) NCERT Pg. No. # 316, 317

167)

NCERT Based

168) NCERT, Pg. # 316,317

169) NCERT-XII, Pg. # 318

170) NCERT-XII, Pg. # 321 (21.4.1)

171) NCERT (XI) Eng. Pg. # 222-223
NCERT (XI) हिन्दी पेज # 223-224

172)

NCERT Pg # 223

173)

NCERT Pg # 223

174) Module no. 3, Pg - 183

175)

NCERT XI Pg. # 212

176)

NCERT XII, Pg. No. 215

177) NCERT Pg. # 214

178)

Generated by Allie

Problem Statement: The question is asking which pair of micronutrients helps during the light phase of photosynthesis, specifically in the reaction that produces oxygen.

Underlying Concept: The light phase of photosynthesis involves splitting water molecules to release oxygen. This process requires certain micronutrients as cofactors in the enzyme complex that drives water splitting. Manganese and Chlorine are essential micronutrients that assist in the oxygen evolution reaction within the photosystem II complex.

Tips and Tricks: Remember that Manganese is crucial in the oxygen-evolving complex, and Chlorine helps stabilize this system. This can help eliminate options that do not include both.

Common Mistakes: Confusing micronutrients that assist in nitrogen fixation (e.g., Molybdenum) or those more related to other phases rather than the light phase of photosynthesis.

Why Other Options Are Incorrect?: Pairs involving Molybdenum or Iron do not participate directly in oxygen evolution during the light phase. Zinc and Chlorine are not directly involved in oxygen evolution; Manganese is key in this process.

179) NCERT Pg. # 214

180)

NCERT Based